Takke

Software Design Document

Takke Employee Salary Calculator

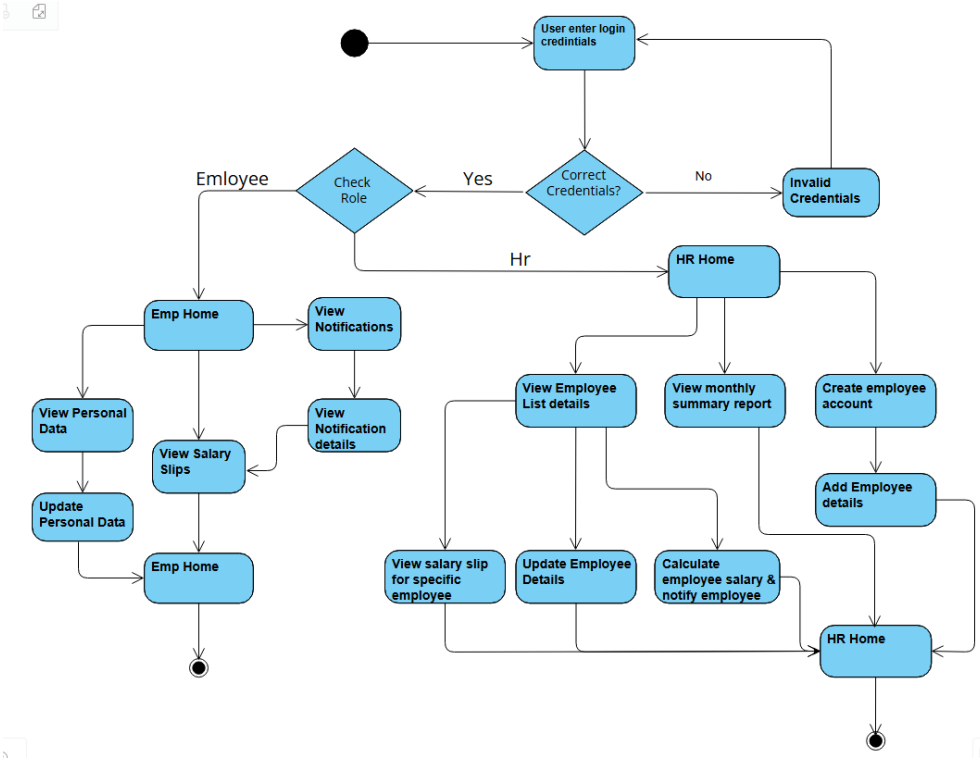
Revision 1.0

Date: (1/18/2026)

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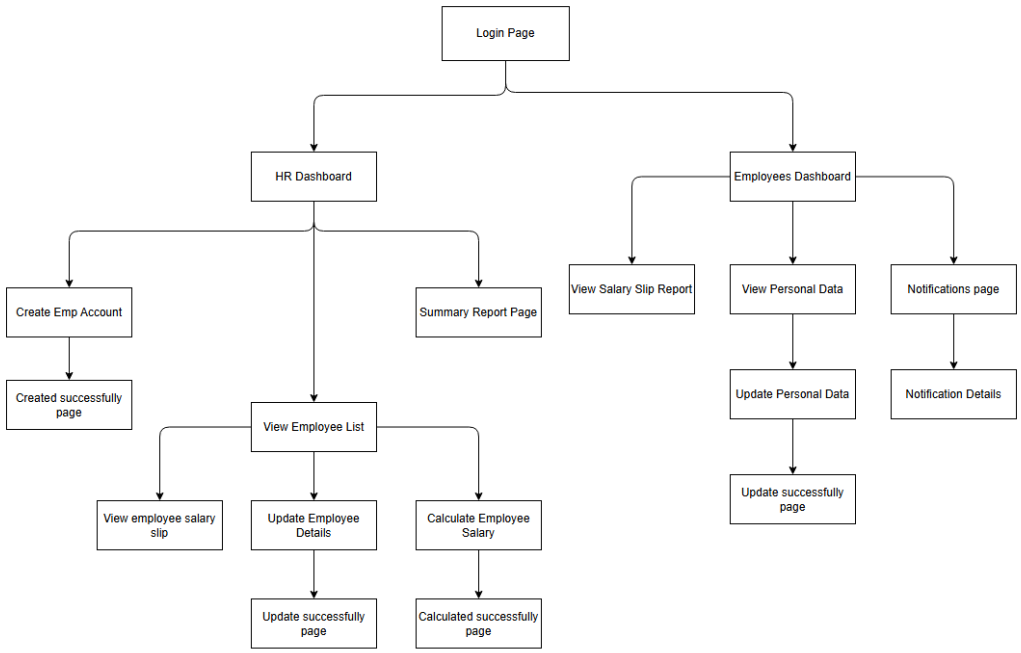
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1. Introduction

1.1 Overview

This is a document called the software design document (SDD) which includes a very large details on the application which is the Takke Employee Salary Calculator (TESC). These details include a plan for building the application which helps us to know what the application is for and how we will build the application.

1.2 Purpose

The purpose of this SDD document is to look at components of the Takke Employee Salary Calculator such as architecture, design, development, implementation, and requirements details which provide a full overview and understanding of the application.

1.3 Project Scope

The main purpose of the Takke Employee Salary Calculator is to make the HR employee's work easier by converting the salary calculation to be fully automated instead of using papers and writing reports by the HR employees, so the application will help HR to add employee, calculate monthly salaries, and generate salary slips.
The goal of this application is to make HR work easier by reducing the manual reporting, increasing productivity and accuracy, and reducing human error which meets the overall vision of any organization.

1.4 Target Audience

The target audience are:

- Front end developers
- Back-end developers
- Testers
- Users
- Designers
- Stakeholders

The SDD is organized into parts:

1. Introduction
2. Overall Description
3. System Analysis
4. Overall Design
5. Overall Development
6. Development Plan Discussion
7. Testing
8. Implementation and Support

All readers must begin with the introduction part and finish with implementation and support part to make a full overview and understanding.

2. Overall Description

2.1 Problem Statement

- The Problem:
  The current process that is being carried out by the HR department is calculating

employees' salary manually based on basic pay, allowances, bonuses and deductions. Also, they add or view employees' details manually, write reports manually, and create salary slips manually.

- The Gap:
  The current system has issues such as that there is high rate of making errors during the process, the process takes a lot of time to be done, the reports are not secured because they are on paper, the papers can be easily manipulated, and if the employees wants to access their data, they have to go to the HR department. So, the client came to us with a specific need which is to develop a platform that integrates web applications used by the HR department and we will provide him with a web-based platform that makes the work that is being carried out by the HR department easier and faster.
- The Impact:
  The new platform will allow HR department to calculate employees' salaries, add or view employee details, generate reports, and generate salary slips and all those operations will be automated, and the employees will access their data without the need to get them from the HR department. Also, the platform will solve the current issues that we talked about earlier.

2.2 Proposed Solution

2.2.1 Project Idea

- Overview:
  Develop a web-based system that allows HR department to calculate employees' salaries, add or view employee details, generate salaries summary report, and generate salary slips. Also, the system will use some data to do its operations such as employee details, basic pay, allowances, bonuses and deductions.
- Key Features:
  - HR dashboard:
    - Create accounts based on employees' roles.
    - Add employee details (name, department, position).
    - Update employee details (name, department, position).
    - View employee details (name, department, position).
    - Calculate net salary based on basic pay, allowances, bonuses, and deductions.
    - View salary slips for specific employee
    - View salaries summary report.
  - Employees dashboard:
    - view their details
    - view their salary reports
    - receive notification when salary is calculated.

- Report generation: the application will generate reports and generate salary slips reports.

2.2.2 Stakeholders

- HR employees:
  - Role: Primary users of the system.
  - Interest: Create employee accounts and add or view employee details and calculate employee salary correctly.
- Other employees:
  - Role: Secondary users of the system.
  - Interest: View their details, along with their salary reports.
- Project Manager:
  - Role: Manage the project from the start to the completion.
  - Interest: Make sure that everything is going well by delivering the project in the highest quality, within the deadline and they make sure that the project does not exceed the planned budget.
- Development team:
  - Role: Design, development, testing, and deployment.
  - Interest: Good design, successful development, project manager and client satisfaction.
- Compliance Authorities:
  - Role: make sure that the system meets the strict privacy and security regulations and requirements for the employees.
  - Interest: Data protection, ensure privacy and security, and ensure legal compliance.

2.2.3 Actors

- HR employees (They will be also the Admin):
  - Description: Primary users that manage employee's accounts information including adding or viewing employee details and making salary calculations process.
  - Responsibilities:
    - Create employees accounts based on their roles.
    - Add employee details.
    - Update employee details
    - view employee details.
    - Calculate employee salary.
    - View employee salary slip.
    - View salaries summary report.
- Other Employees:
  - Description: Secondary users of the system.

- Responsibilities:
  - View their personal data.
  - View their salary slips.
  - Receive notifications when salary is calculated.

2.3 Product Overview

2.3.1 Functional requirement

You need to specify the functional requirement in detail. Provide a full description of the feature and indicate whether it is of High, Medium, or Low priority. You could also include specific priority component ratings, such as benefit, penalty, cost, and risk (each rated on a relative scale from a low of 1 to a high of 9).

- HR/Admin Management: (High Priority)
  - The system shall allow HR employees to add/update or view employee details that includes (name, department, position).
  - The system shall allow HR to create accounts for the users based on the user role.
  - The system shall allow HR employees to calculate employees' salaries based on specific criteria:
    - Basic pay
    - Allowances (e.g., transport, housing)
    - Bonuses and Deductions (e.g., tax, insurance).
  - The system shall allow HR to view a report for financial purposes which includes a summary of all employees salaries.
  - The system shall allow HR to view a salary slip reports for specific employee.
- Employees Management: (Medium Priority)
  - The system shall allow employees to view their personal data.
  - The system shall allow employees to view their final salary slip.
  - The system shall allow employees to update their personal information.
  - The system shall allow employees to receive in-app notification when salary is calculated.
- Login and Authentication: (High Priority)
  - The system shall allow employees to login securely based on user role.
- Reporting: (High Priority)
  - The system shall generate monthly summary salaries report.
  - The system shall generate salary slips.
  - The system shall allow customization of a report based on criteria such as month, year, and department.

2.3.2 Nonfunctional Requirement (Quality Attributes)

2.3.2.1 Reliability

- The system should generate the reports within 5 seconds.
- The system should make the calculations 100% correct.
- The login process should be done within 3 seconds.

2.3.2.2 Availability

- The system should have a 99.9% uptime.
- The system should have at most 0.1% downtime.

2.3.2.3 Security

- The system should restrict unauthorized access.
- The system should have a clear permission for users.
- The system should use data encryption techniques.

2.3.2.4 Maintainability

- The system designs, application, calculations, logic, and database should be well documented.
- The system should be developed to accept future updates.

2.3.2.5 Portability

- The system should be accessible on different browsers.
- The system should be responsive as users can open it on different desktop screen sizes.

2.4 Areas of Risk

Risk	Effect	Priority	Penalty	Cost	Mitigation
Design – unclear requirements	It can lead to misunderstanding and perform a bad design in the user interface or in the database which leads to user or performance issues.	High	9	8	Set clear requirements at the start of the project and use the best methodology for the project.
Design – unachievable design.	It can make the project exceed the deadline as the design should be remade.	High	7	9	Give design team clear requirements with instructions.
Development – Security risks.	It could lead to unauthorized access or unauthorized use of the system.	High	10	9	Implement strong security techniques such as encryption, access control

					lists, and secure authentication.
Development – Unclear requirements.	It may lead to developing the desired functionality in a wrong way which could overrun time and budget.	High	9	8	Set clear requirements at the start of the project and use the best methodology for the project.
Testing – not enough test cases.	Not enough test cases may lead to not testing some essential parts in the system.	Medium	6	8	Give the testing team instructions about how they chose the test cases for example (each functionality should have 3 test cases at least).
Project exceed time and budget.	Project exceeds the time and budget that has been set.	High	10	10	Use suitable methodology along with setting a realistic deadline and cost.
Adoption challenges.	Difficulty getting used to the new system process.	Low	4	5	Train the employees to make them used to the new system.

3. System Analysis

3.1 Use case Diagrams (per functionality)

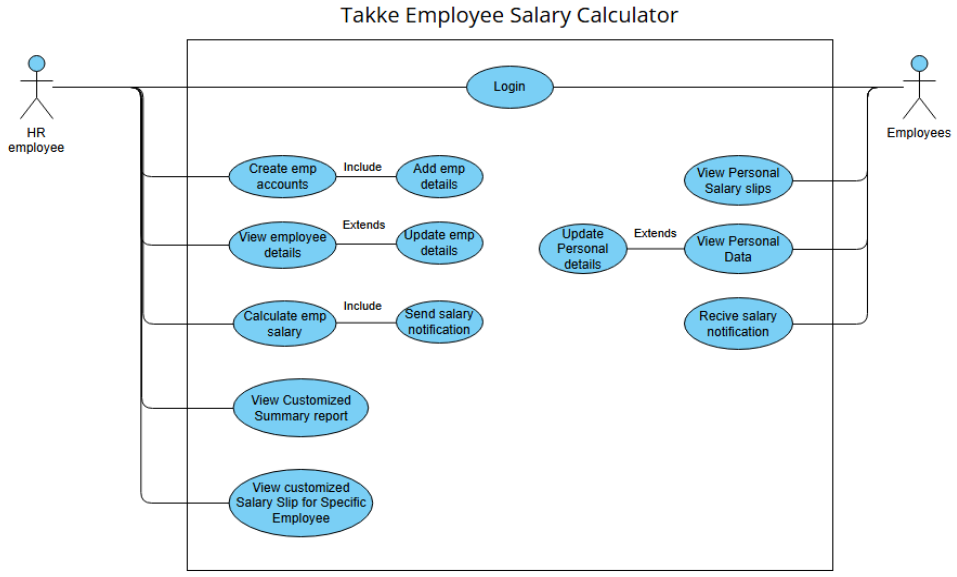


Figure 1: Use Case Diagram

A use case diagram to explain the main functions and interactions between the users and the system which we represented the goals of the interactions, organizing functional requirements, modeled the flow of events.

3.2 Activity Diagram

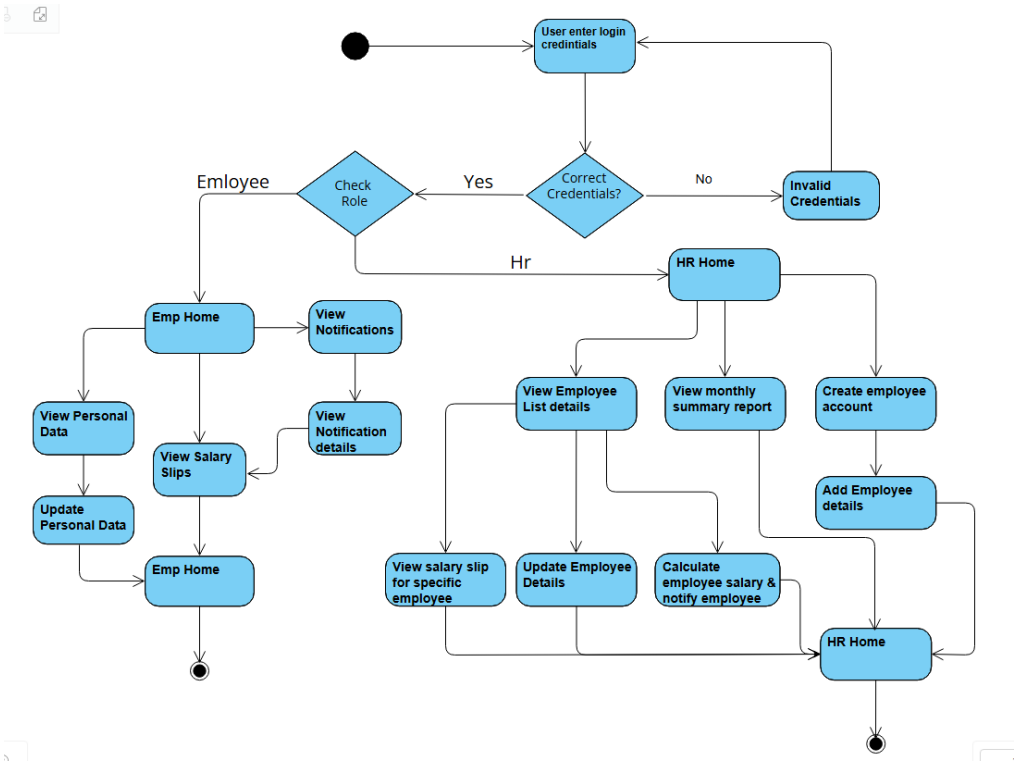


Figure 2: Activity Diagram

An activity diagram was designed in order to show the activities that the system performs and in which flow, as we helped the people from the business and development to understand the process and behaviour of the system. As we clearly defined the logic of the whole system, show the steps performed, and identify the workflow.

3.3 Component Diagram

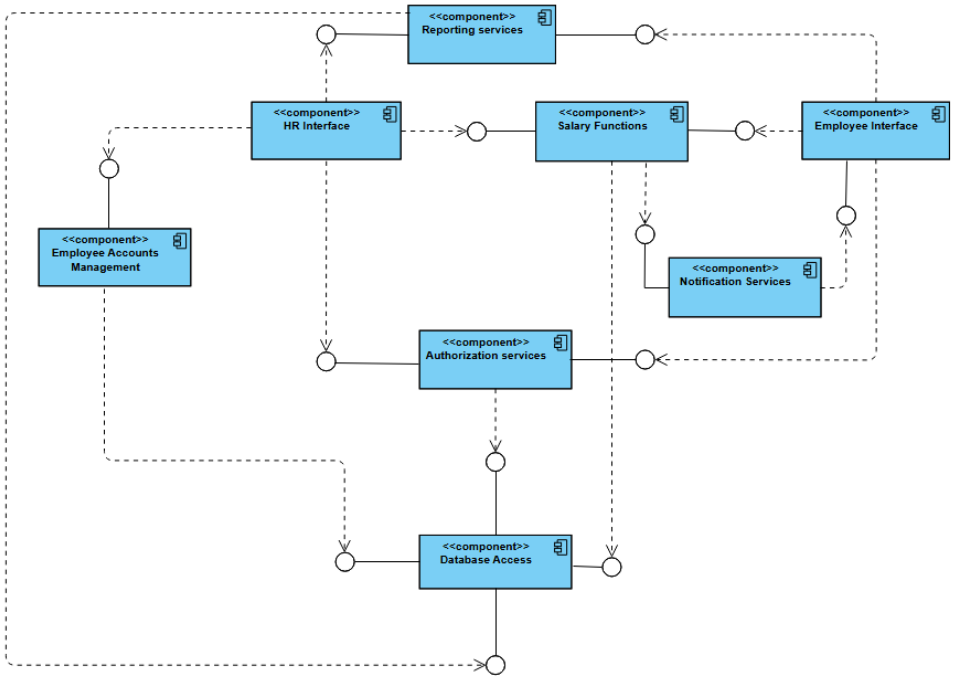


Figure 3: Component Diagram

A component diagram was designed in order to break the system under development to different levels of components and functionalities, as each component is responsible for one goal in the system and how it interact with other components, as our goal here to see the relationship between the system's components.

3.4 Class Diagram

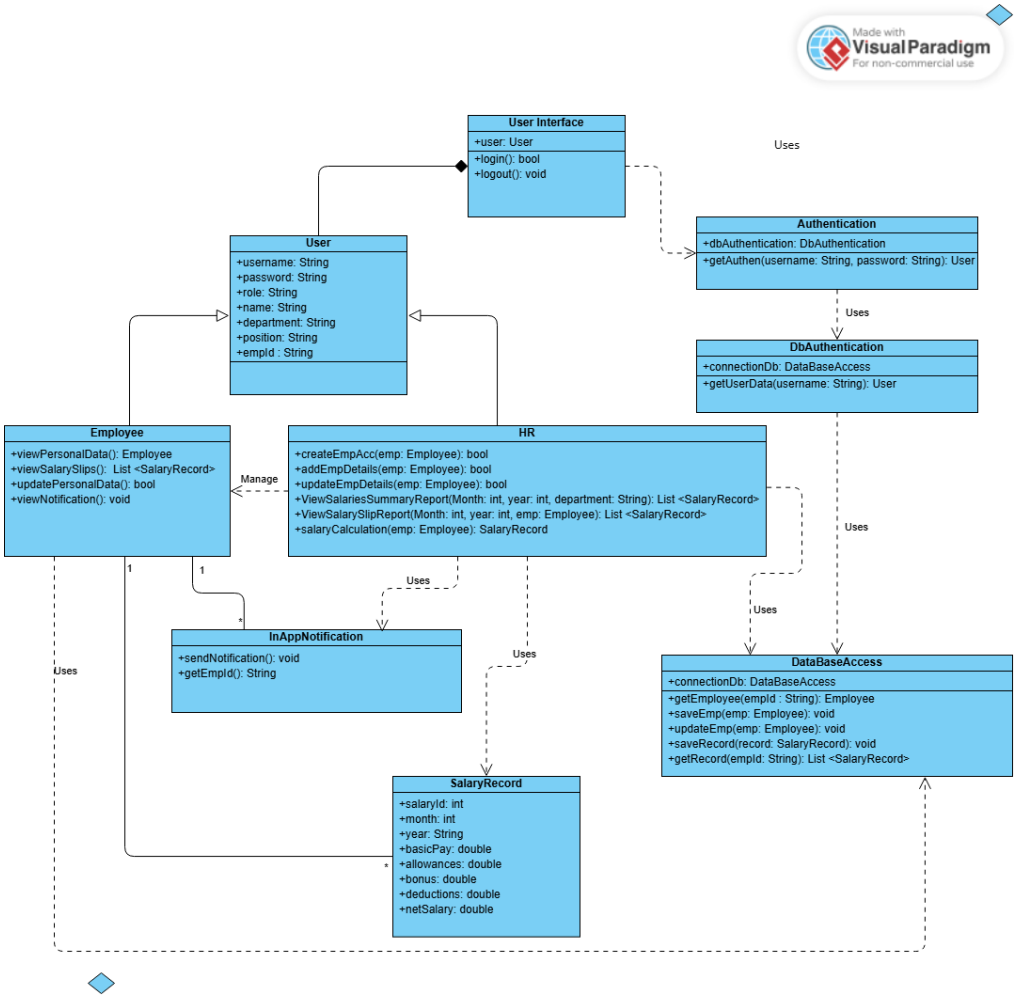


Figure 4: Class Diagram

A class diagram was designed to have a structured look on the system, by converting the system into classes that are part of the system including the attributes and relationship, we were able to understand data models, the underlying things of the system, created specific parts, understand the types of system's parts.

3.5 ERD

The original design:

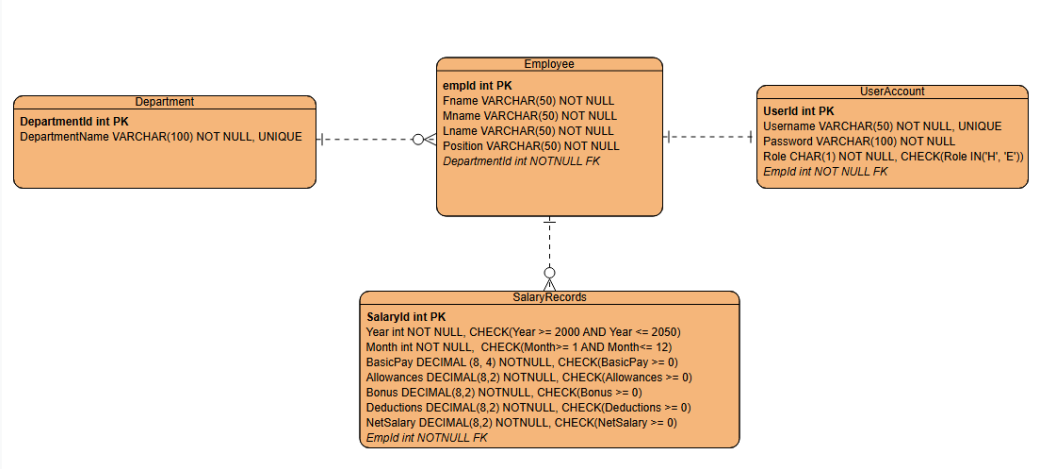


Figure 5: ERD (Original)

But because of the development platform limitations we created another version:

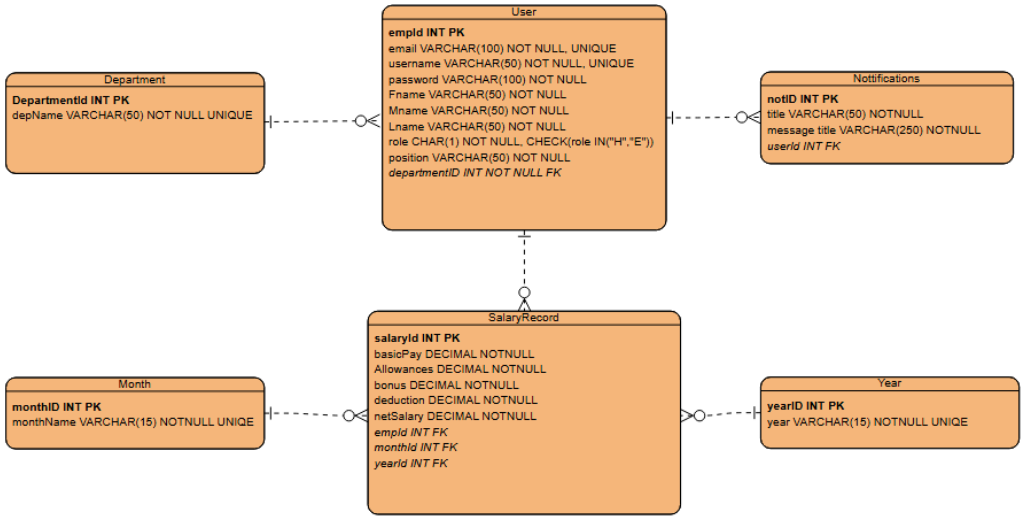


Figure 6: ERD (new version)

A ERD was designed to have a better look on the data storage and databases in order to understand the database design including tables and attributes.

4. Overall Design

4.1 Sitemap

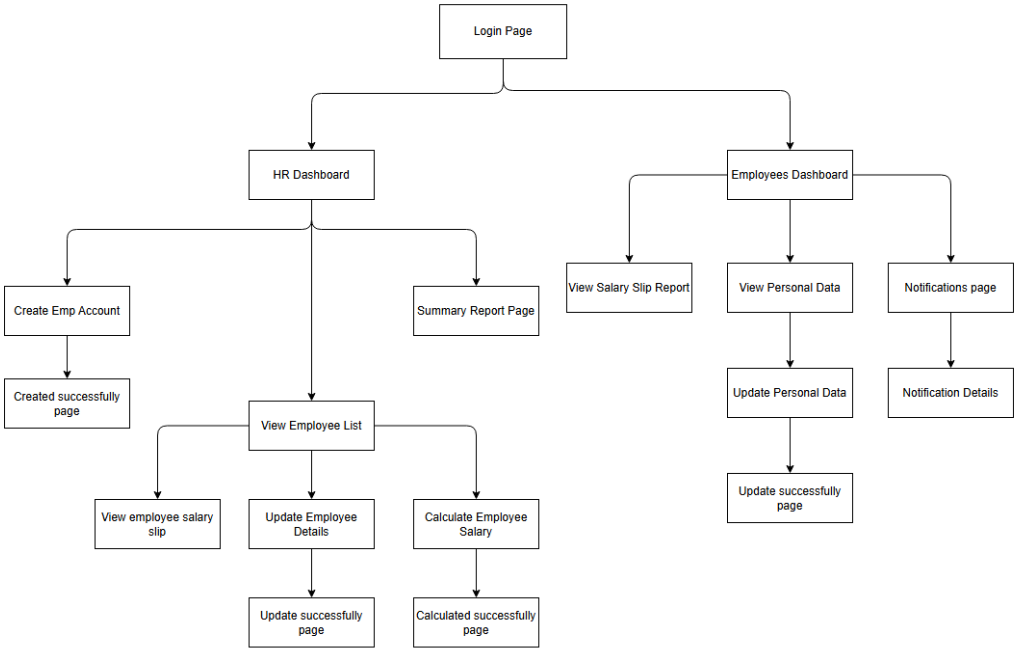


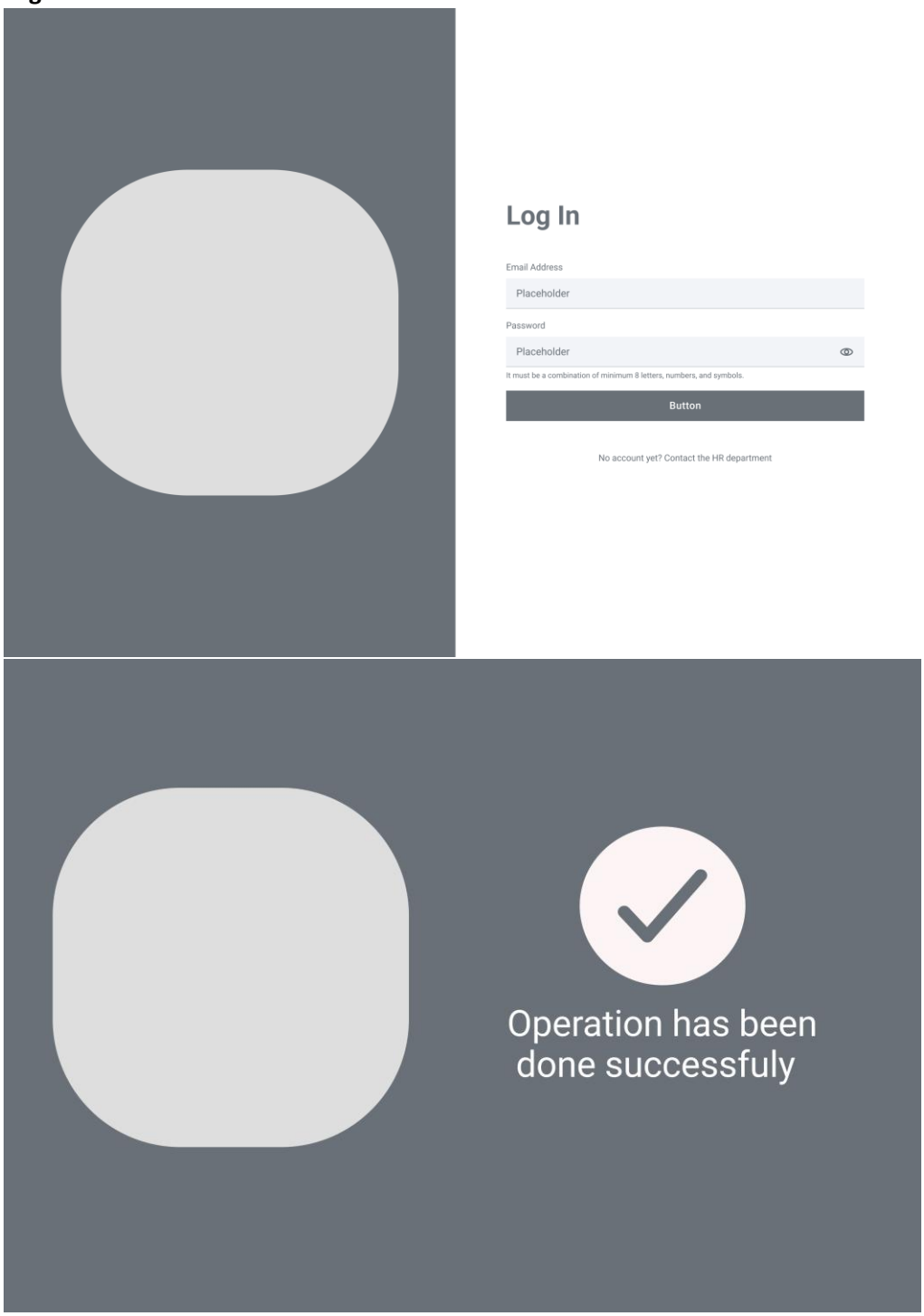
Figure 7: Sitemap diagram

A Sitemap diagram was designed in order to plan all the pages that are in the application and to see how they will be connected to each other.

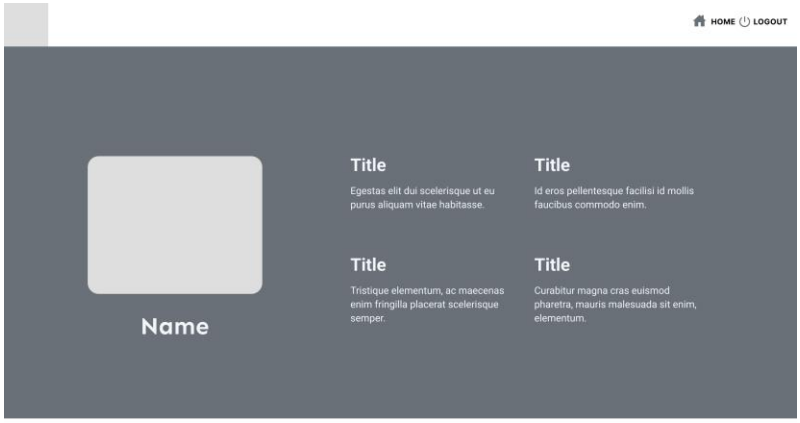
4.2 Wireframes

Figure 8: Wireframe Figures

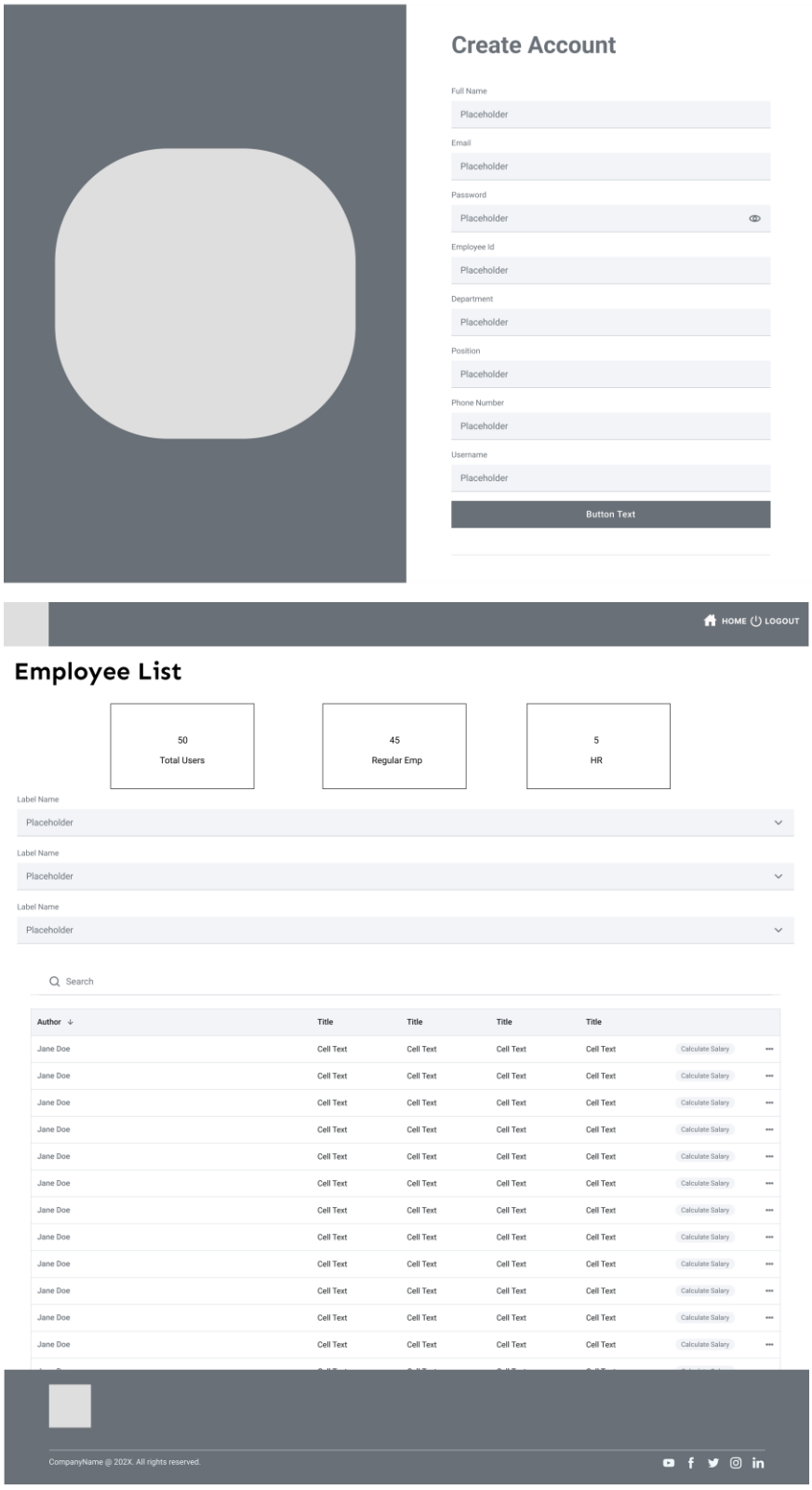
Login and Feedback:

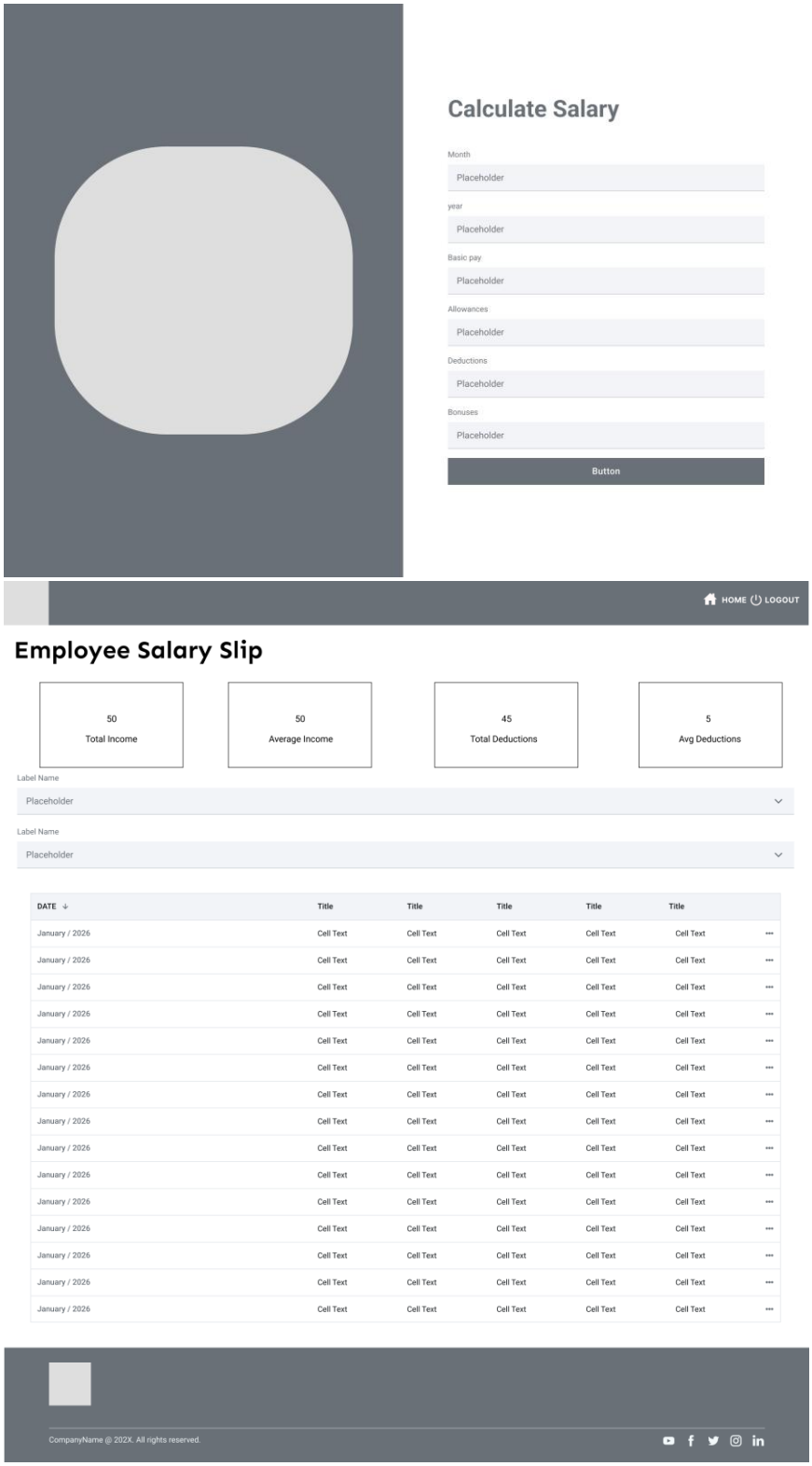


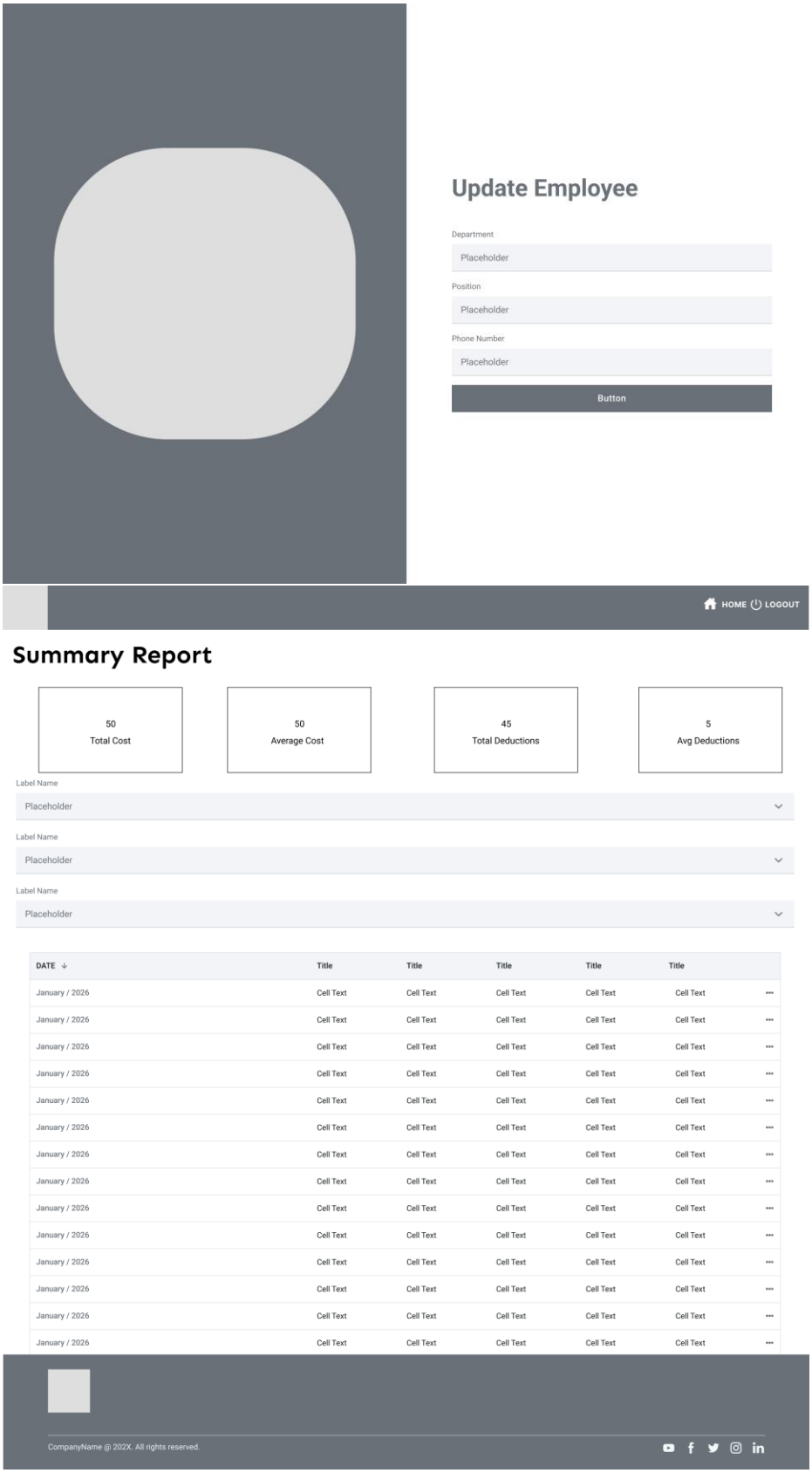
HR Employees:



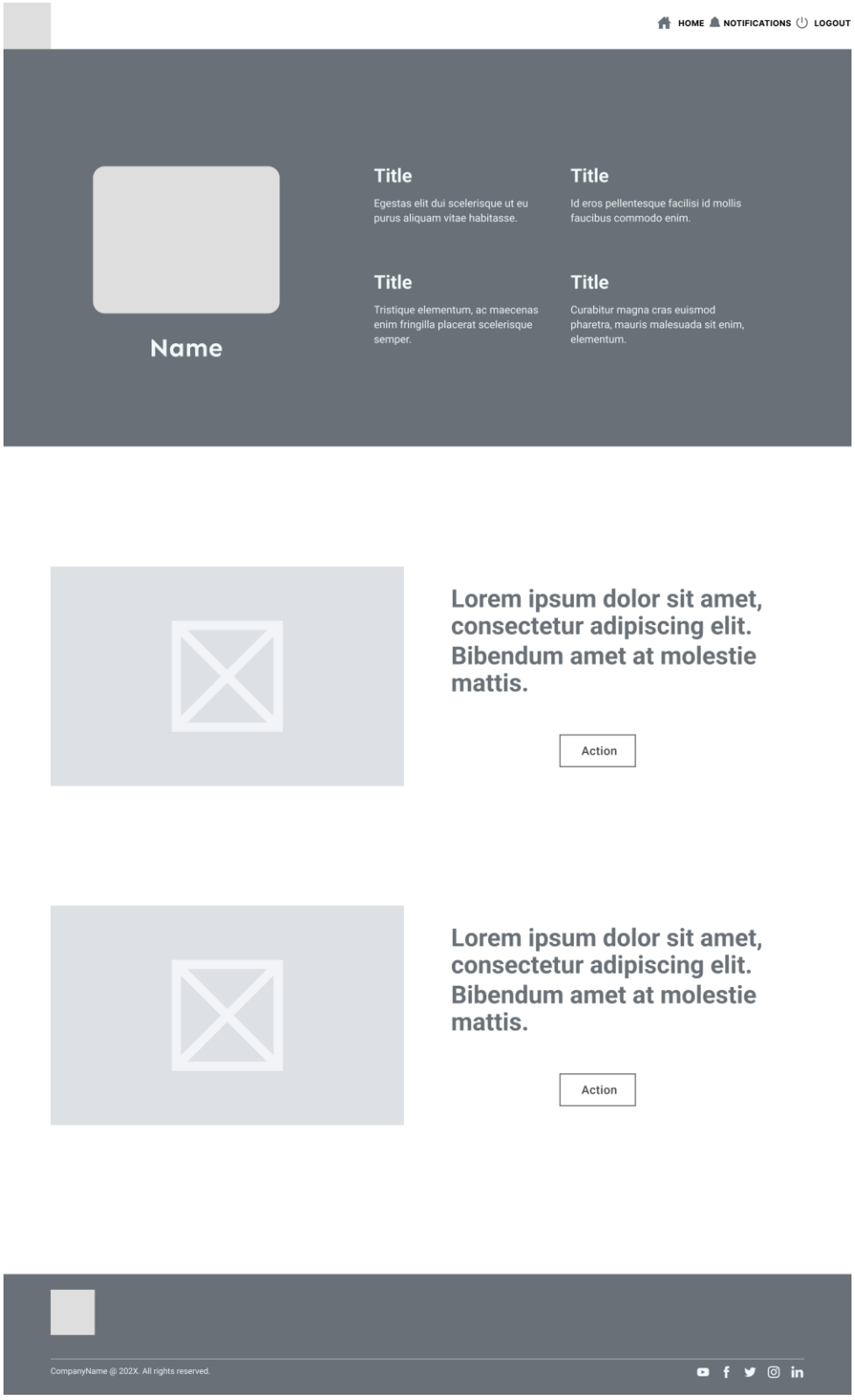


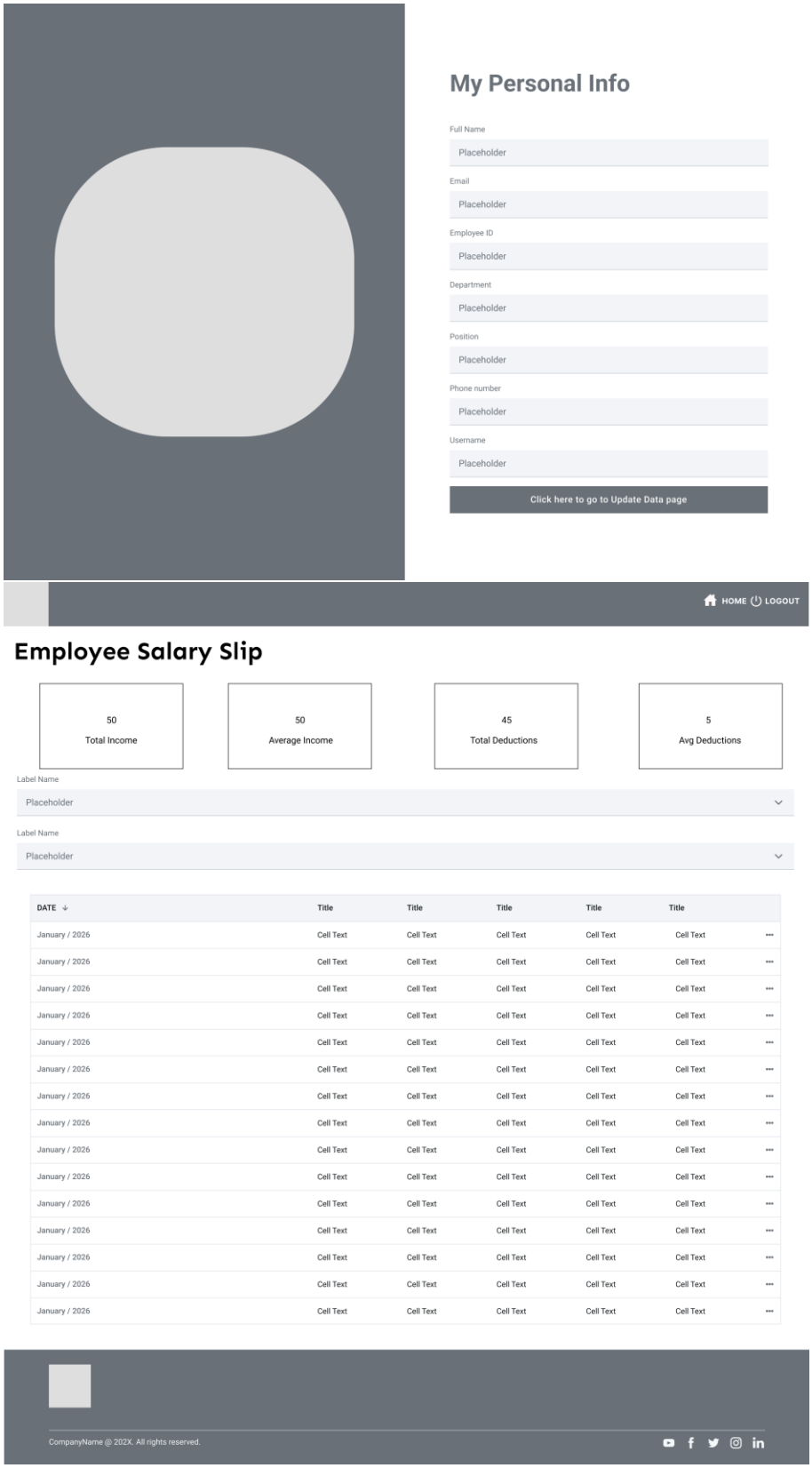

Create Account
Full Name
Placeholder
Email
Placeholder
Password
Placeholder
Employee Id
Placeholder
Department
Placeholder
Position
Placeholder
Phone Number
Placeholder
Username
Placeholder
Button Text
HOME
LOGOUT
Employee List
50
Total Users
45
Regular Emp
5
HR
Label Name
Placeholder
Label Name
Placeholder
Label Name
Placeholder
Search
Author
Title
Title
Title
Title
Jane Doe
Cell Text
Calculate Salary
CompanyName @ 202X. All rights reserved.

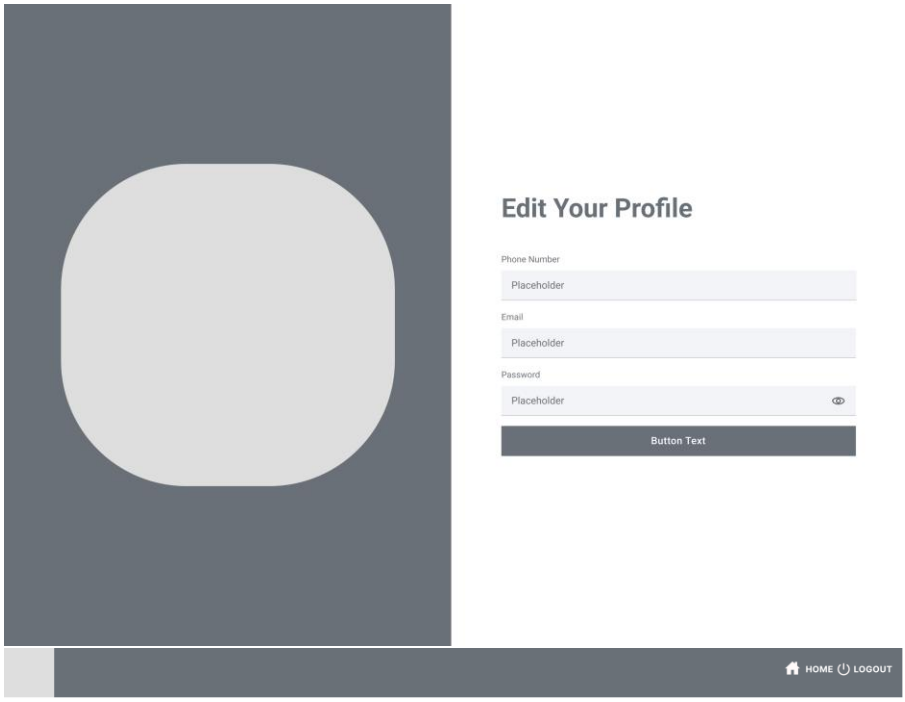

Calculate Salary
Month
Placeholder
year
Placeholder
Basic pay
Placeholder
Allowances
Placeholder
Deductions
Placeholder
Bonuses
Placeholder
Button
HOME
LOGOUT
Employee Salary Slip
50
Total Income
50
Average Income
45
Total Deductions
5
Avg Deductions
Label Name
Placeholder
Label Name
Placeholder
DATE
Title
Cell Text
January / 2026
CompanyName @ 202X. All rights reserved.


Update Employee
Department
Placeholder
Position
Placeholder
Phone Number
Placeholder
Button
HOME
LOGOUT
Summary Report
50
Total Cost
50
Average Cost
45
Total Deductions
5
Avg Deductions
Label Name
Placeholder
Label Name
Placeholder
Label Name
Placeholder
DATE
Title
Title
Title
Title
Title
January / 2026
Cell Text
CompanyName @ 202X. All rights reserved.

Regular Employees:



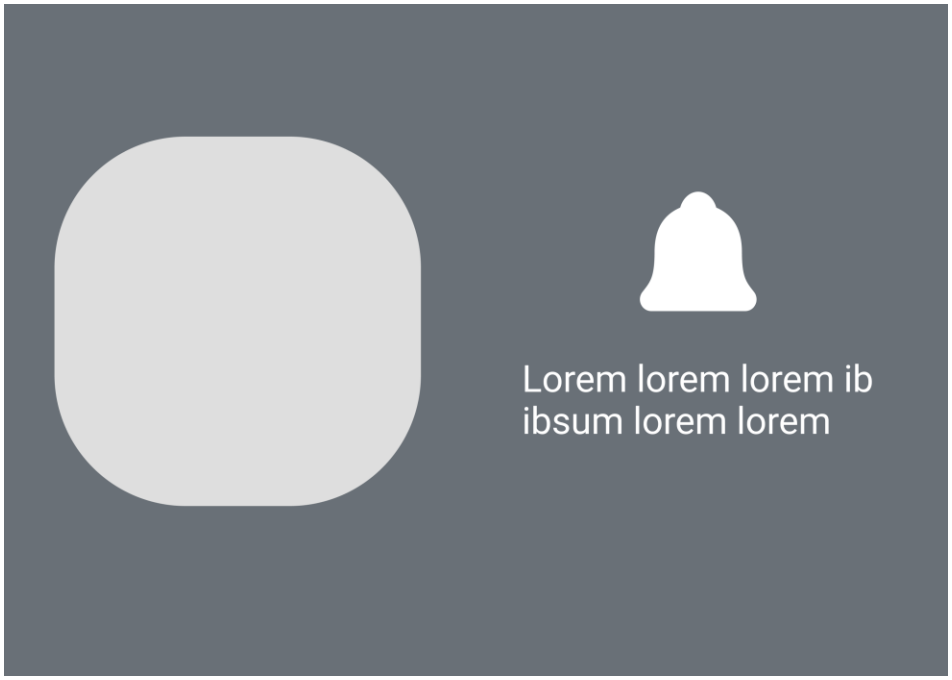

My Personal Info
Full Name
Placeholder
Email
Placeholder
Employee ID
Placeholder
Department
Placeholder
Position
Placeholder
Phone number
Placeholder
Username
Placeholder
Click here to go to Update Data page
HOME
LOGOUT
Employee Salary Slip
50
Total Income
50
Average Income
45
Total Deductions
5
Avg Deductions
Label Name
Placeholder
Label Name
Placeholder
DATE
Title
Cell Text
January / 2026
CompanyName @ 202X. All rights reserved.



Employee's Notifications

DATE	Title
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
Jane Doe	Cell Text
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Jane Doe	Cell Text



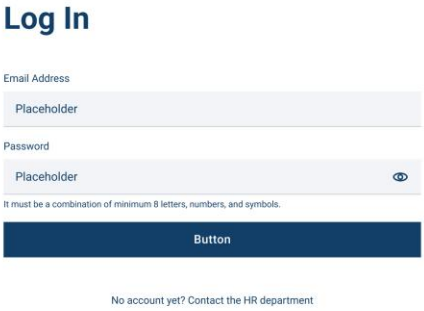

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ibsum lorem lorem

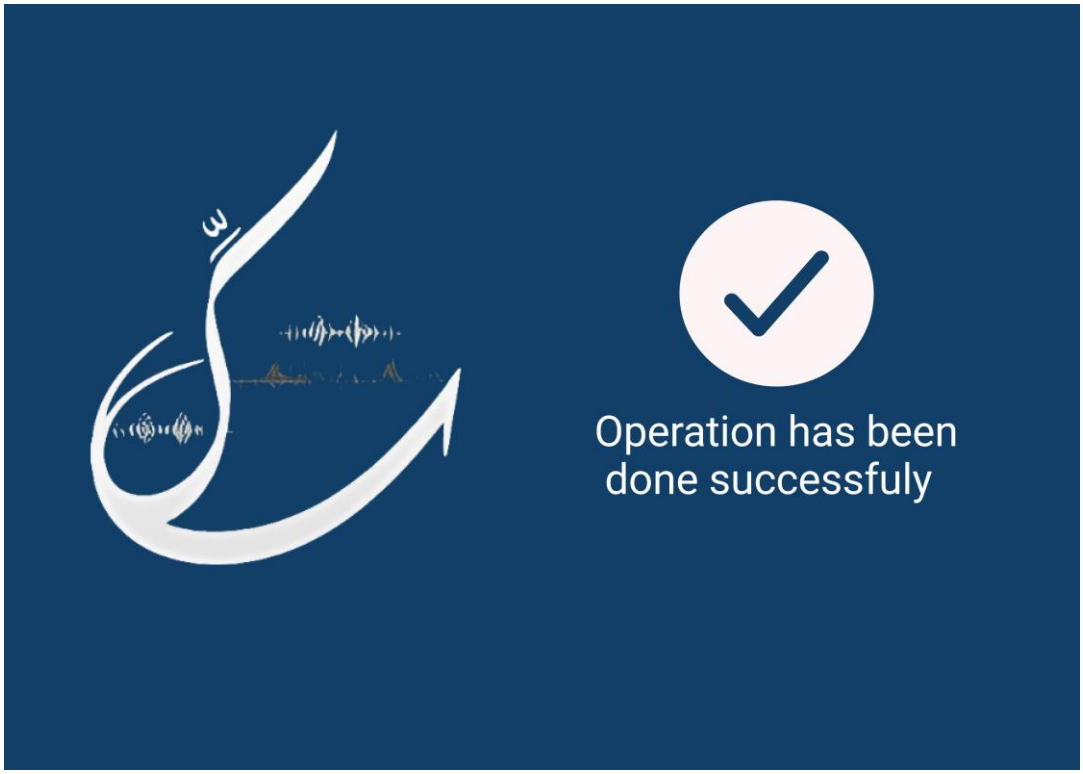
4.3 Mockups

Figure 9: Mockup Figures

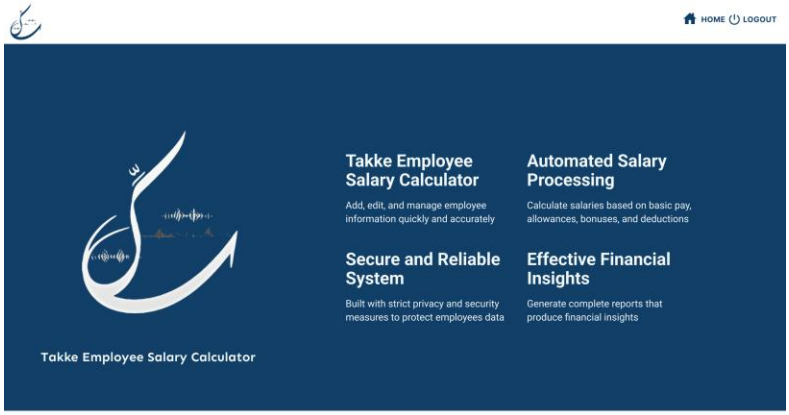
Login and feedback:







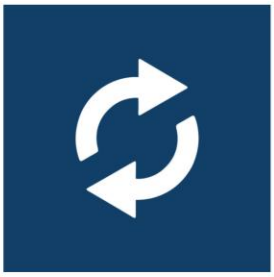
HR:





Add New Employee To The System

Action



View, edit, and update employee information

Action



View a financial report and insights

Action





Create Account

Full Name
Placeholder
Email
Placeholder
Password
Placeholder
Employee Id
Placeholder
Department
Placeholder
Position
Placeholder
Phone Number
Placeholder
Username
Placeholder
Button Text



Employee List

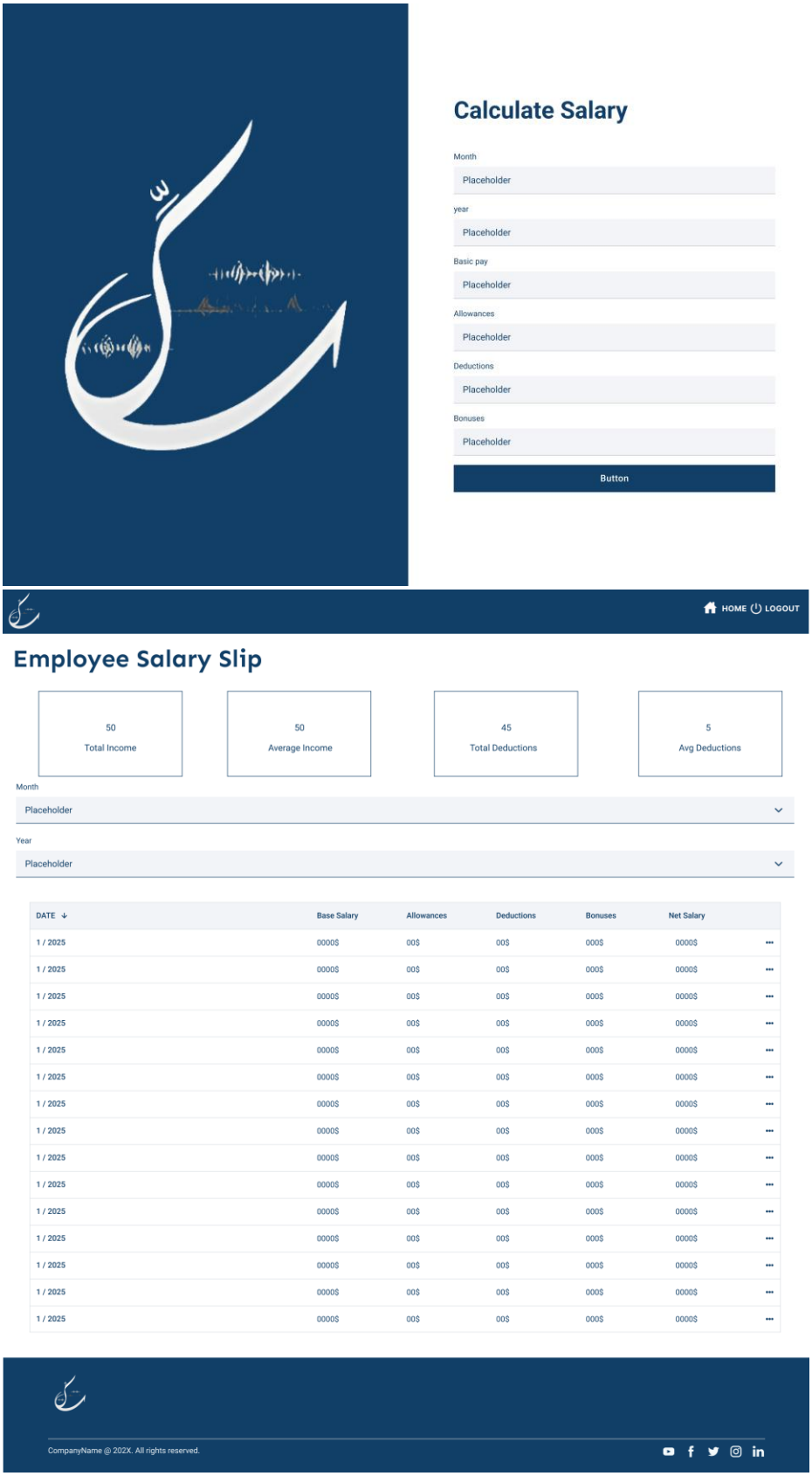
50	45	5
Total Users	Regular Emp	HR

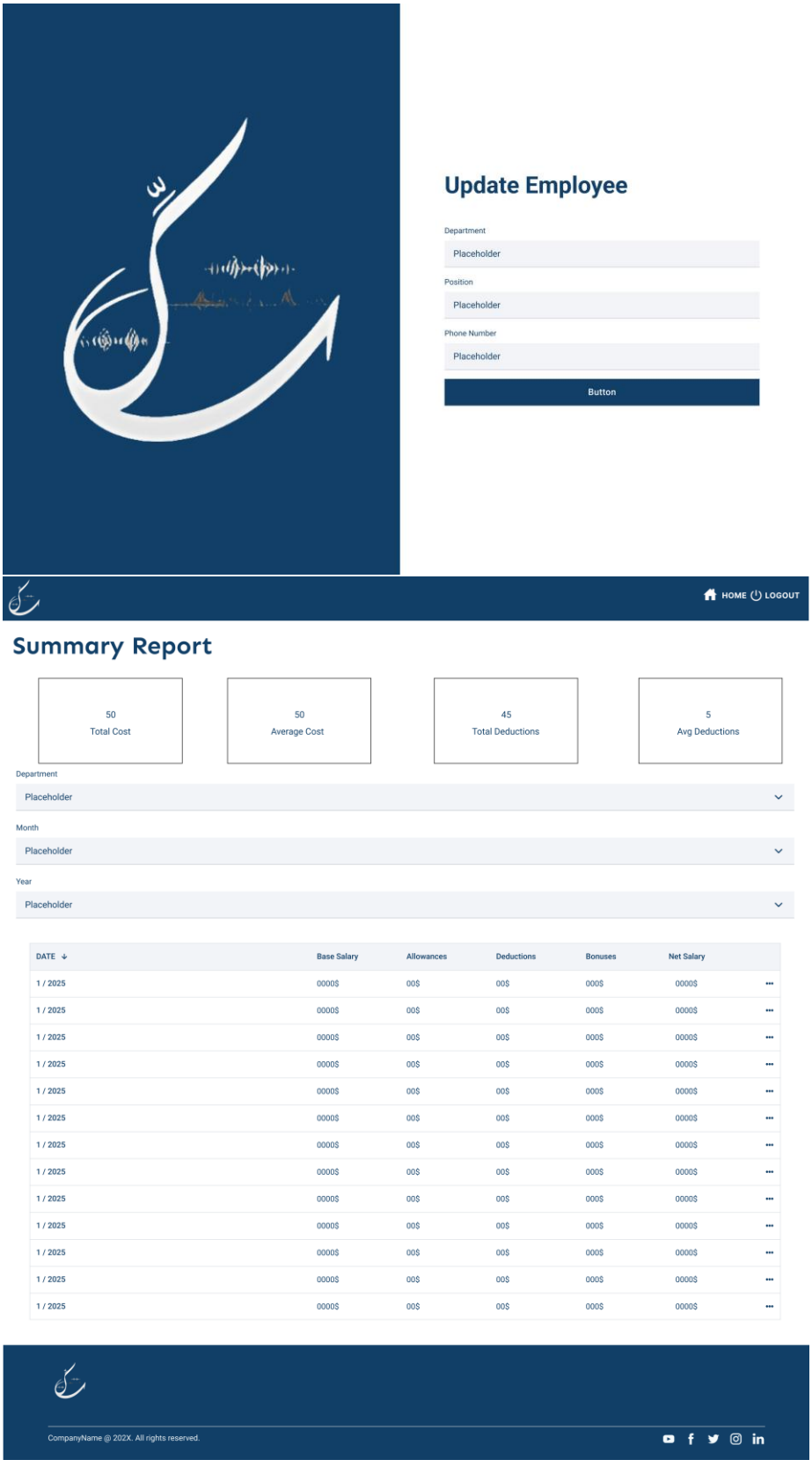
Department
Placeholder

Search

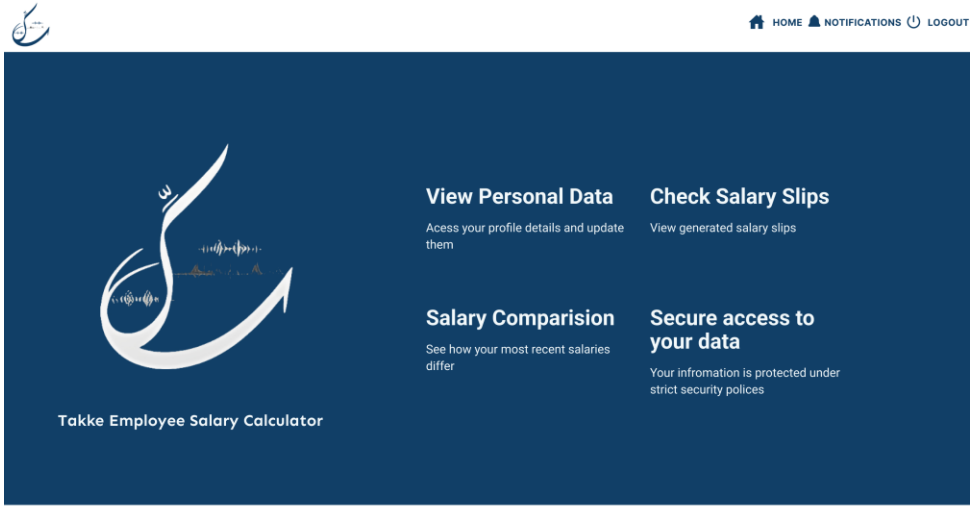
Name	Email	ID	Department	Position	
Jane Doe	Jane@gmail.com	0-2002	Operations	Position	Calculate Salary
Jane Doe	Jane@gmail.com	0-2002	Operations	Position	Calculate Salary
Jane Doe	Jane@gmail.com	0-2002	Operations	Position	Calculate Salary
Jane Doe	Jane@gmail.com	0-2002	Operations	Position	Calculate Salary
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Jane Doe	Jane@gmail.com	0-2002	Operations	Position	Calculate Salary




Calculate Salary
Month
Placeholder
year
Placeholder
Basic pay
Placeholder
Allowances
Placeholder
Deductions
Placeholder
Bonuses
Placeholder
Button
HOME
LOGOUT
Employee Salary Slip
50
Total Income
50
Average Income
45
Total Deductions
5
Avg Deductions
Month
Placeholder
Year
Placeholder
DATE
Base Salary
Allowances
Deductions
Bonuses
Net Salary
1 / 2025
0000$
00$
000$
CompanyName @ 202X. All rights reserved.


Update Employee
Department
Placeholder
Position
Placeholder
Phone Number
Placeholder
Button
HOME
LOGOUT
Summary Report
50
Total Cost
50
Average Cost
45
Total Deductions
5
Avg Deductions
Department
Placeholder
Month
Placeholder
Year
Placeholder
DATE
Base Salary
Allowances
Deductions
Bonuses
Net Salary
1 / 2025
0000$
00$
000$
CompanyName @ 202X. All rights reserved.

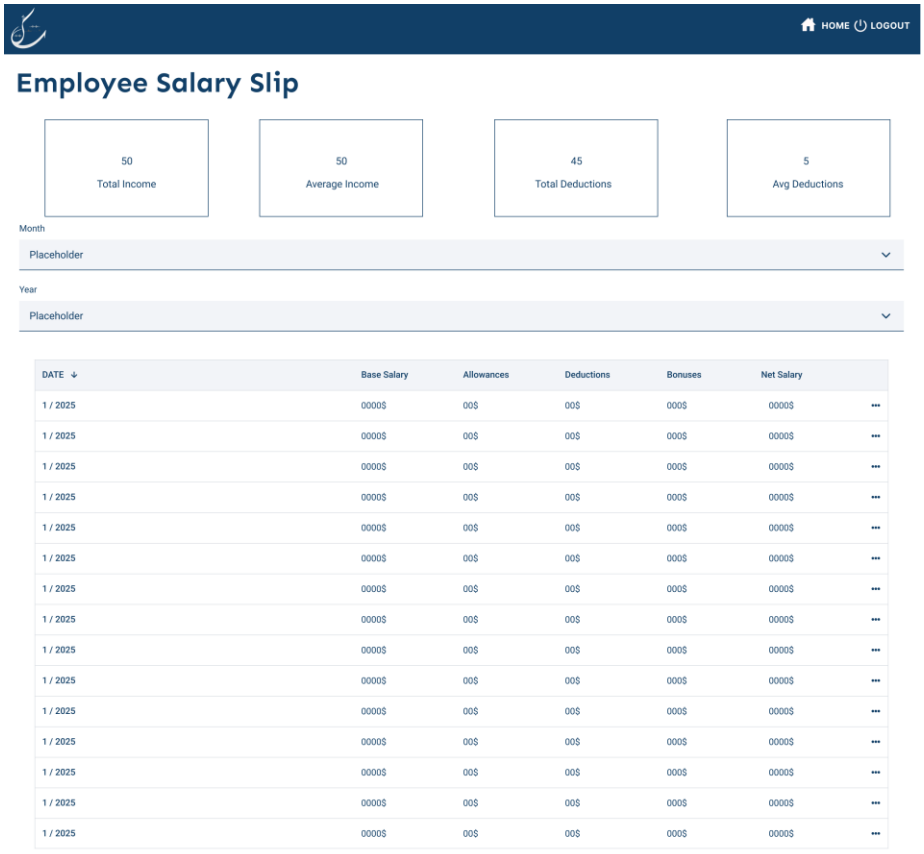
Employees:


HOME
NOTIFICATIONS
LOGOUT
Takke Employee Salary Calculator
View Personal Data
Acess your profile details and update them
Check Salary Slips
View generated salary slips
Salary Comparision
See how your most recent salaries differ
Secure access to your data
Your infromation is protected under strict security polices


View & Update your personal information
Action

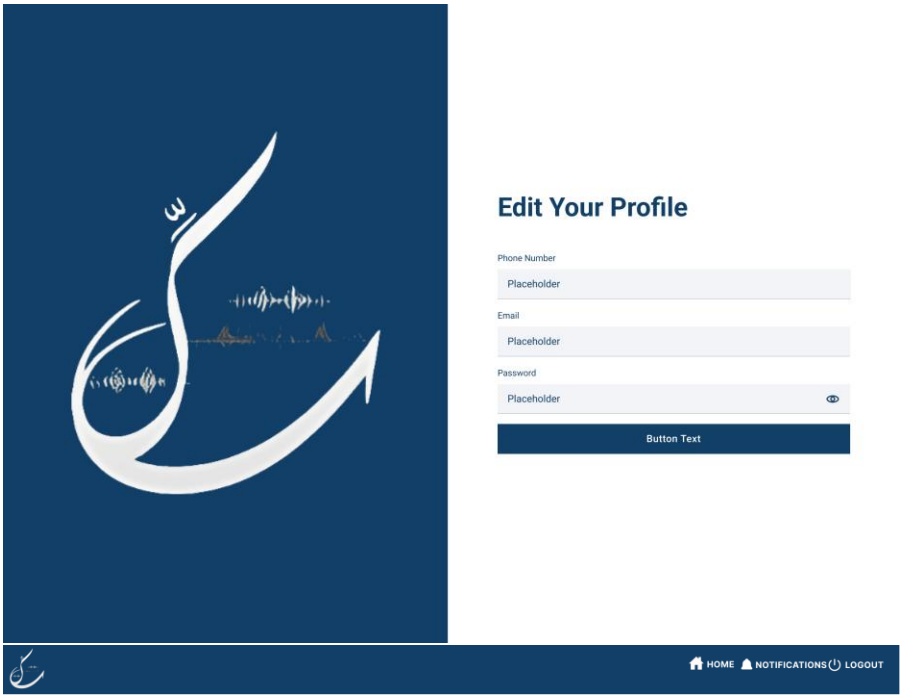

Access your monthly salary slips
Action


CompanyName @ 202X. All rights reserved.


HOME
LOGOUT
Employee Salary Slip
50
Total Income
50
Average Income
45
Total Deductions
5
Avg Deductions
Month
Placeholder
Year
Placeholder
DATE
Base Salary
Allowances
Deductions
Bonuses
Net Salary
1 / 2025
0000$
00$
000$


CompanyName @ 202X. All rights reserved.


My Personal Info
Full Name
Placeholder
Email
Placeholder
Employee ID
Placeholder
Department
Placeholder
Position
Placeholder
Phone number
Placeholder
Username
Placeholder
Click here to go to Update Data page



Employee's Notifications

Title	Date
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026
Your Salary has been calculated	2 / 2026



5. Overall Development

5.1 The preferable Development Tool, Technique, and Software Development Methodology

There are many development tools, techniques, and software development methodologies. In this section we will show the selected ones and list their advantages and disadvantages.

For the development techniques, I choose no-code RAD because it supports quick development iterations, changes are welcomed, have a high flexibility, and it increases the customer satisfaction rate. But it has disadvantages like it is not suitable for application that has complex logic.

For the design tool, I chose Figma as it is a cloud design tool that helps designers to fastly make wireframes and mockups, and it was chosen because it helps me to design all the pages visually and on saved on cloud which converted the requirements into visual designs.

For the design methodology, I choose sitemap, wireframe, and mockups in order to understand the design of the UI of the overall system.

For the development tool, I chose Adalo as it uses drag-and-drop components, and I chose it because it allows fast and rapid development and getting quick feedback, and it is suitable for building an application that supports business processes like HR dashboards, and because it reduces time and cost of the development. It has some advantages like the limit on customization, and we have less control on the back end, and it is not ideal for big systems, but our system is not very big.

For the development methodology, I choose RAD because the TESC does not have a fixed requirement and they may change, and it matches the tool used for development. It has advantages such as it is very flexible, more customer involvement, changes are welcomed, and it Is fast. It also has disadvantages such as it needs highly skilled developers, always want to ask the customer and the customer may not be available, and it needs good management of iterations.

- ## Development Plan Discussion

a. Development Review

i. Peer review feedback

We made a feedback session after we finished the system design, and we needed to collect feedback about problem statements, project idea, proposed solution, functional

and non-functional requirements, system analysis diagrams, wireframes, and mockups, and we concluded to collect feedback that most of them were about the functional requirements.

1. Reporting: Add report customization like filtering the report based on the date which will help HR employees and regular employees to view the report in a specific date.
2. Notifications or Alerts: Add some kind of notification or alert system to employees when HR calculate their salary.
3. Employee data: Add a feature where a regular employee can update their own personal data.
4. Compare Salaries: Remove the feature “Compare the last two months salaries”, because it is not useful to make it a full requirement, as report customization and filters allow employees to compare salaries.

ii. Interpret the peer review feedback

The peer review feedback was collected and considered then we implemented it in the Software Design Document and in the application itself.

The reporting feedback was considered and the feature was only displaying a report without any customization or filters, so we concluded to realizing that without filtering the report, the employee should go through all the records in the report to find a specific record, so this feature improve the overall user experience, as when an employee wants to view records for a specific date, they can only choose that date and not go through all the records to find it.

The notification or alerts feedback was also considered, and we realized that regular employees need an alert if their salary has been calculated, as without these alerts, employees has no idea that their salary has been calculated, so we added in-app notification feature to notify the employees that their salary has been calculated, which improves the communication.

Also, the employee personal data feedback was considered, and we implement it as soon as we heard it, because it is reasonable that the account owner should be able to update their personal data which improves data accuracy and security, and satisfy real-world application needs.

The last feedback we collected was removing the compare the last two months salaries feature, so we considered it, and looked on how the feature work, we realized that this feature does not add any value when there is filters on the reports, as filters allow employees to compare their records already, so this feature was removed to reduce unnecessary features.

All of these edits made us change some things in the system analysis design diagrams

including use case, activity, component, and class diagrams, as we removed the comparing salary feature from all the diagrams, added the notification or alert feature, and we added “Allow employee to update their personal data features”.

Also, some features were added or modified, as we improved the design of the application, improved the overall navigation flow, allowed HR to see a specific employee report, added feedback page when action is done, and enhance filtering report by choosing the department.

iii. Identify and evaluate new opportunities

After collecting the feedback, we identified new opportunities that were not considered before during the first stages, as most of these opportunities came during the development phase or even before it, and some of these opportunities were implemented and some of them were considered as future enhancement.

We identified the first opportunity which is improving the visual design of the home pages, report screens, and some other screens, as it became clear that the initial designs could be better for the user, so we evaluated this opportunity and it became clear that this opportunity should be implemented, as we implemented it, and saw a difference in the previous designs and the new designs which improved the overall user experience.

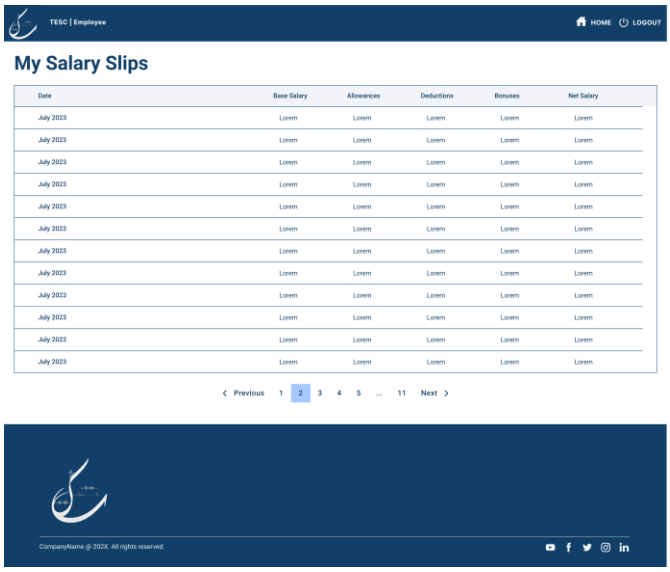


Figure 10:The Salary Slip Screen Before



Figure 11: The Salary Slip Screen After

Another opportunity was changing the application flow, as the original navigation made some issues to the home pages, as we needed to add more visuals to the home page which made the design complexity high, so we redesigned the flow of the application in order to reduce the design complexity in the home pages which improves the usability and efficiency.

Also, there was a very important opportunity which is adding a feedback or confirmation page after completing actions like update or create and including calculating salaries or updating information. So, we implemented this opportunity due to many factors such as that the user needs a confirmation that an action has been done successfully or unsuccessfully, improve clarity, improve usability, and improve user satisfaction.

Also, another opportunity came to our minds which is allowing HR to view salary reports for a specific employee and not only viewing salary reports for all employees, as this opportunity was implemented because it is what the real HR employees should be able to, as whatever the HR employee can do without the system, they should be able to do them easier with the system.

Also, there is an opportunity came to our minds while developing the report customization feature which is the HR should not only filter the report based on the date, but also they should filter the report based on the departments and it was implemented because HR sometimes want to view reports for a specific employee and

as we said before whatever the HR employee can do without the system, they should be able to do them easier with the system.

There are opportunities that were considered as a future enhancement or improvement such as recording which HR employee calculated each salary, and it was considered as a future improvement because the system is not complex now and we don't want to go outside the scope of the system.

Also, exporting reports as PDF files was considered as an opportunity but we added it the future enhancement because of the platform limitations, time and cost constraints that are connected to Adalo.

b. Application Development

i. Database

The database used is the built-in Adalo database, which is a relational database system where data are stored in a collection that act like tables in traditional databases and there are relationships between these collections.

This database was selected because it aligns well with the chosen approach, which is RAD approach because it supports fast development, fast and easy maintenance, and it does not require any complex coding.

So overall, the Adalo database is very suitable for Takke employee salary calculator because it allows for fast collection creation and management, fast and easy relationship creation and management, support filtering, and very suitable for applications that does not require complex setup like the Takke employee salary calculator.
The database structure was designed with multiple collections as these are the main collections in the system:

- Users:
  - Email, text
  - Password, password
  - Username, text
  - EmpId, text
  - First Name, text
  - Middle Name, text
  - Last Name, text
  - isHR, Boolean
  - phone number, number
- SalaryRecord:
  - Basic Pay, number
  - Allowances, number
  - Bonuses, number



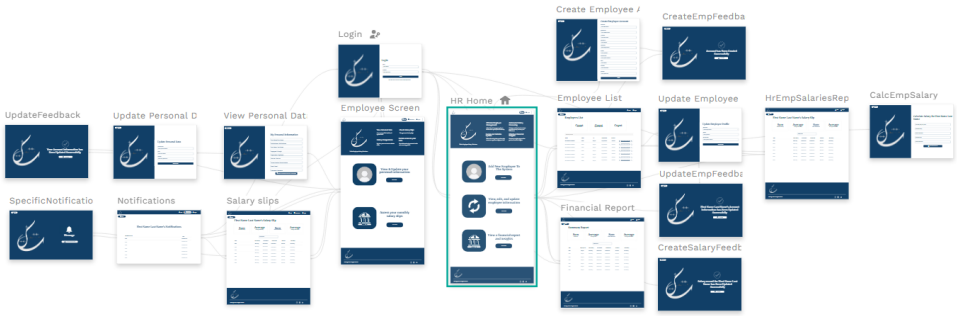
    - Deductions, number
    - Net Salary, number
- Month:
    - Month, text
- Year:
    - Year, text
- Department:
    - DepName, text
- Notifications:
    - Title, text
    - Message, text
- Relationships:
    - SalaryRecord - users (one-to-many, one user, many salary record)
    - Department – users (one-to-many, many users, one department)
    - Notifications – users (one-to-many, many notifications, one users)
    - Month – SalaryRecord (one-to-many, one month, many salaryrecords)
    - Year – SalaryRecord (one-to-many, one year, many salaryrecords)

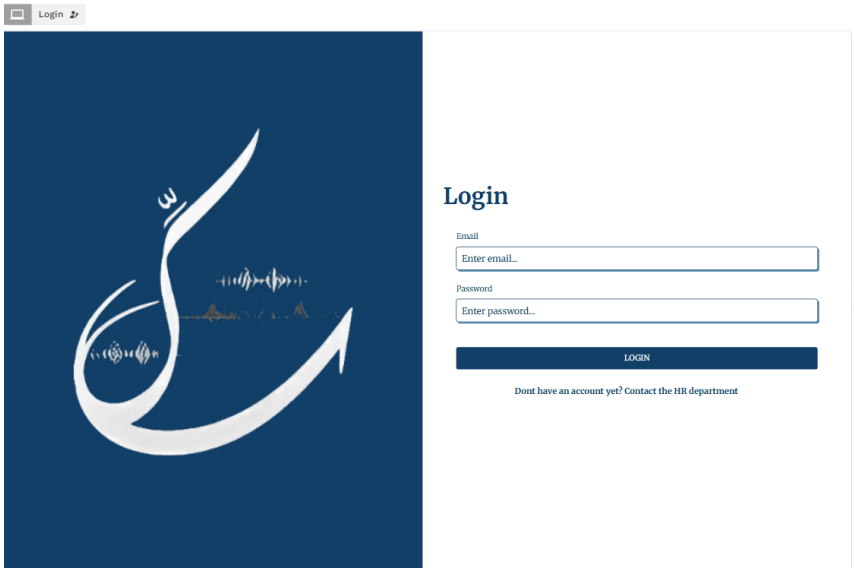
The month, year, and department collection has data that will never be touched, and they were created in order to make the report customization achievable, as due to Adalo limitations, if these were created as attributes inside the SalaryRecord collection, the report customization would not be achievable.

ii. Screenshots of your application

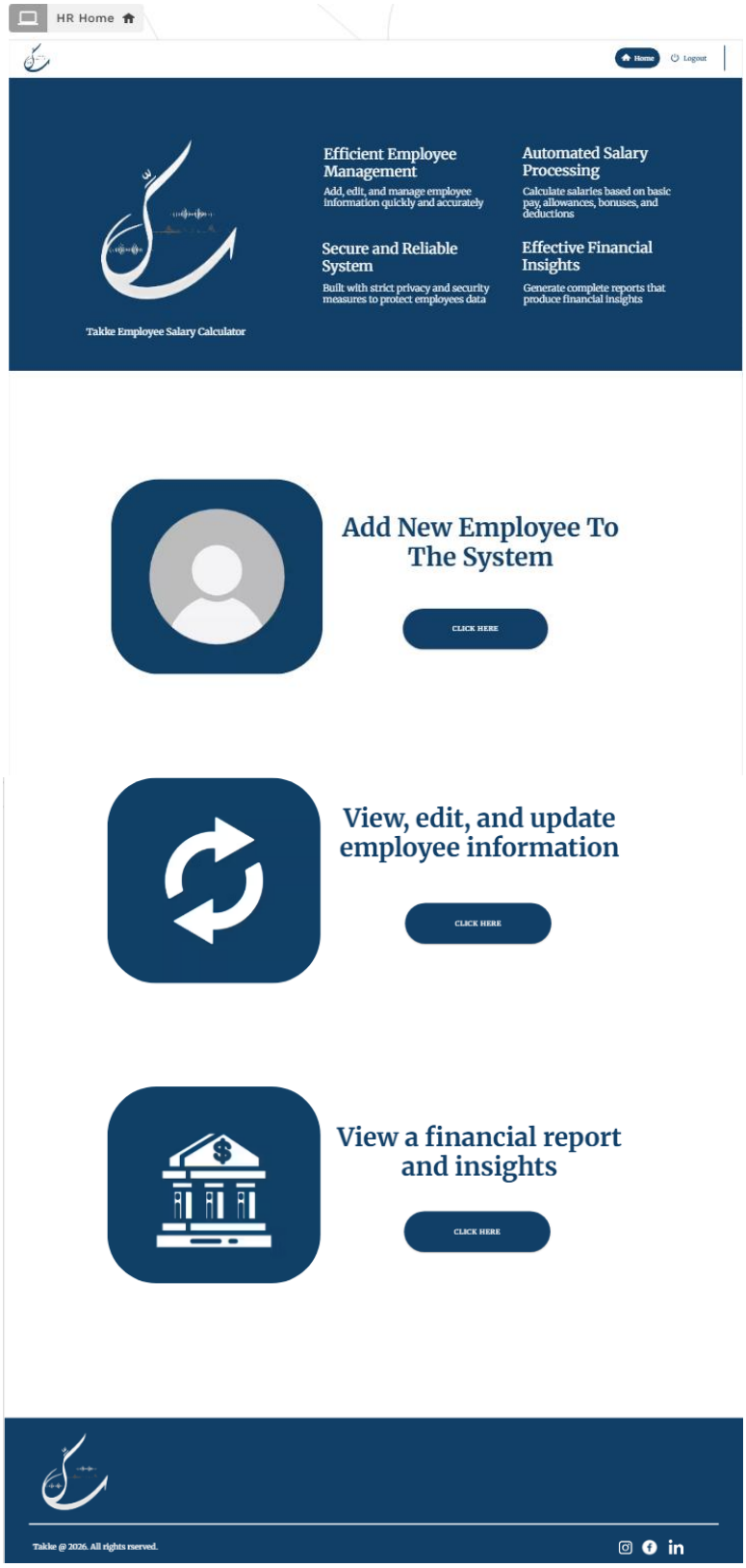
Application Link: TESC Adalo
Overview Video: TESC Overview 23110581

Figure 12: Application Screenshots





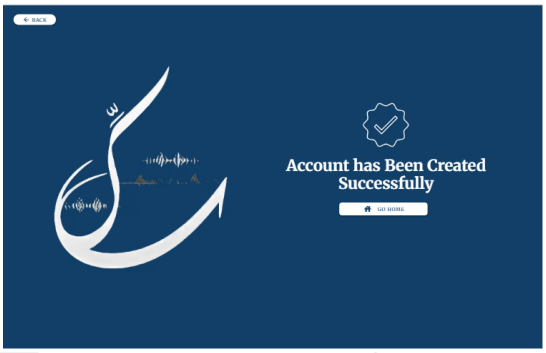
HR:


HR Home
Home
Logout
Takke Employee Salary Calculator
Efficient Employee Management
Add, edit, and manage employee information quickly and accurately
Automated Salary Processing
Calculate salaries based on basic pay, allowances, bonuses, and deductions
Secure and Reliable System
Built with strict privacy and security measures to protect employees data
Effective Financial Insights
Generate complete reports that produce financial insights
Add New Employee To The System
CLICK HERE
View, edit, and update employee information
CLICK HERE
View a financial report and insights
CLICK HERE
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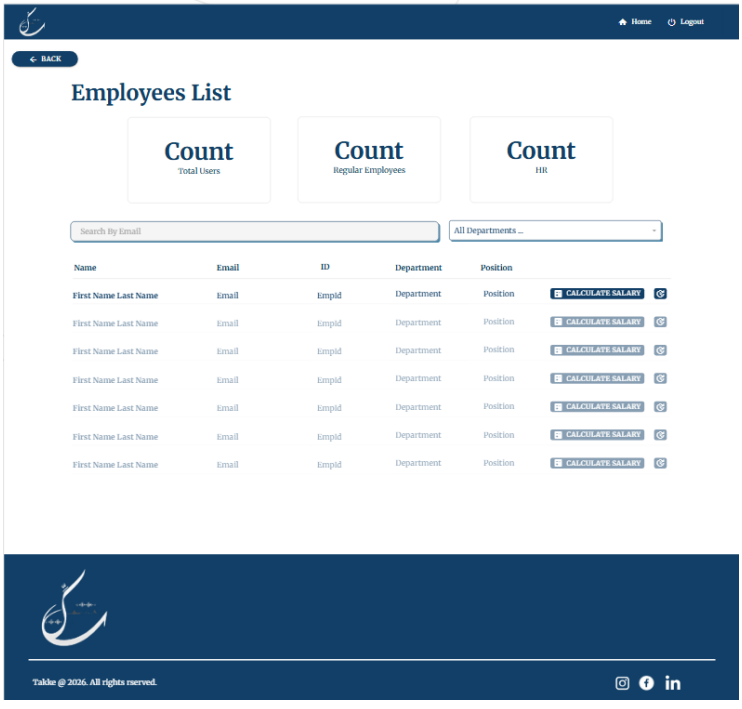
Create Employee Account

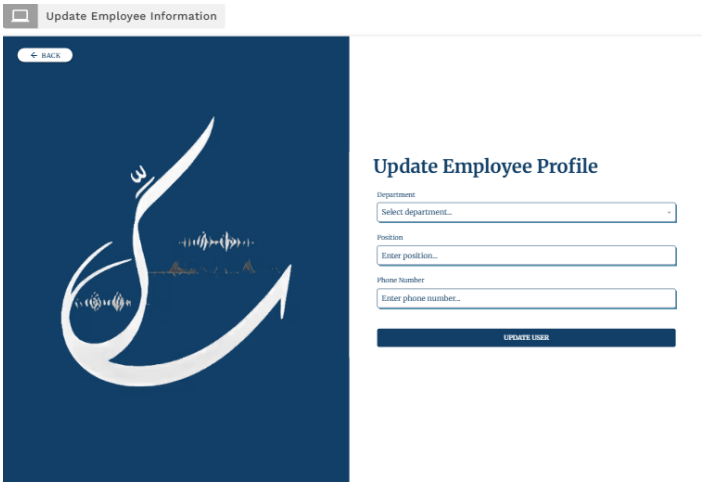


CreateEmpFeedback

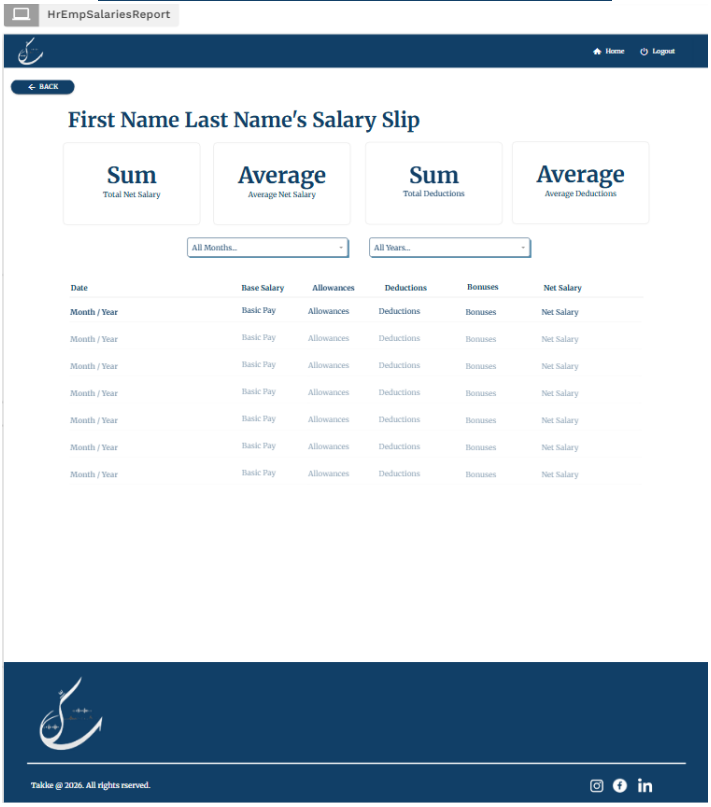


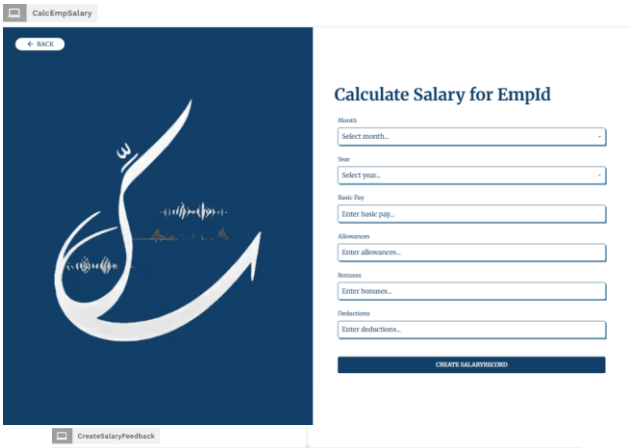
Employee List

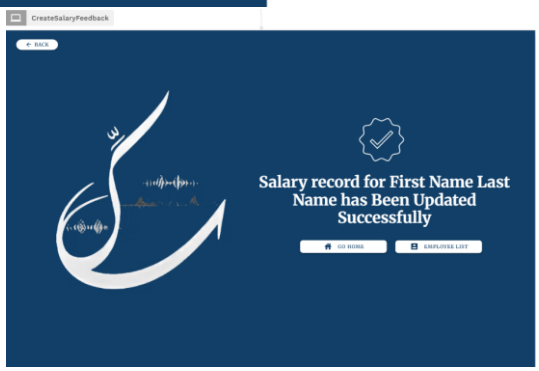


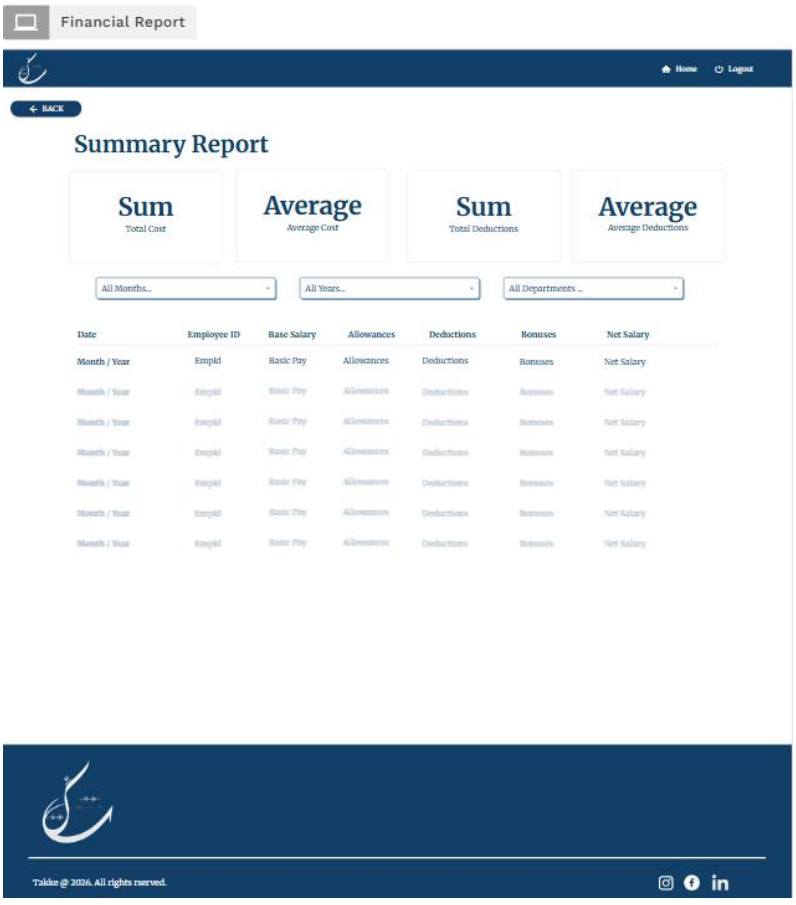

Update Employee Information
BACK
Update Employee Profile
Department
Select department...
Position
Enter position...
Phone Number
Enter phone number...
UPDATE USER


UpdateEmpFeedback
First Name Last Name's Account Information has Been Updated Successfully

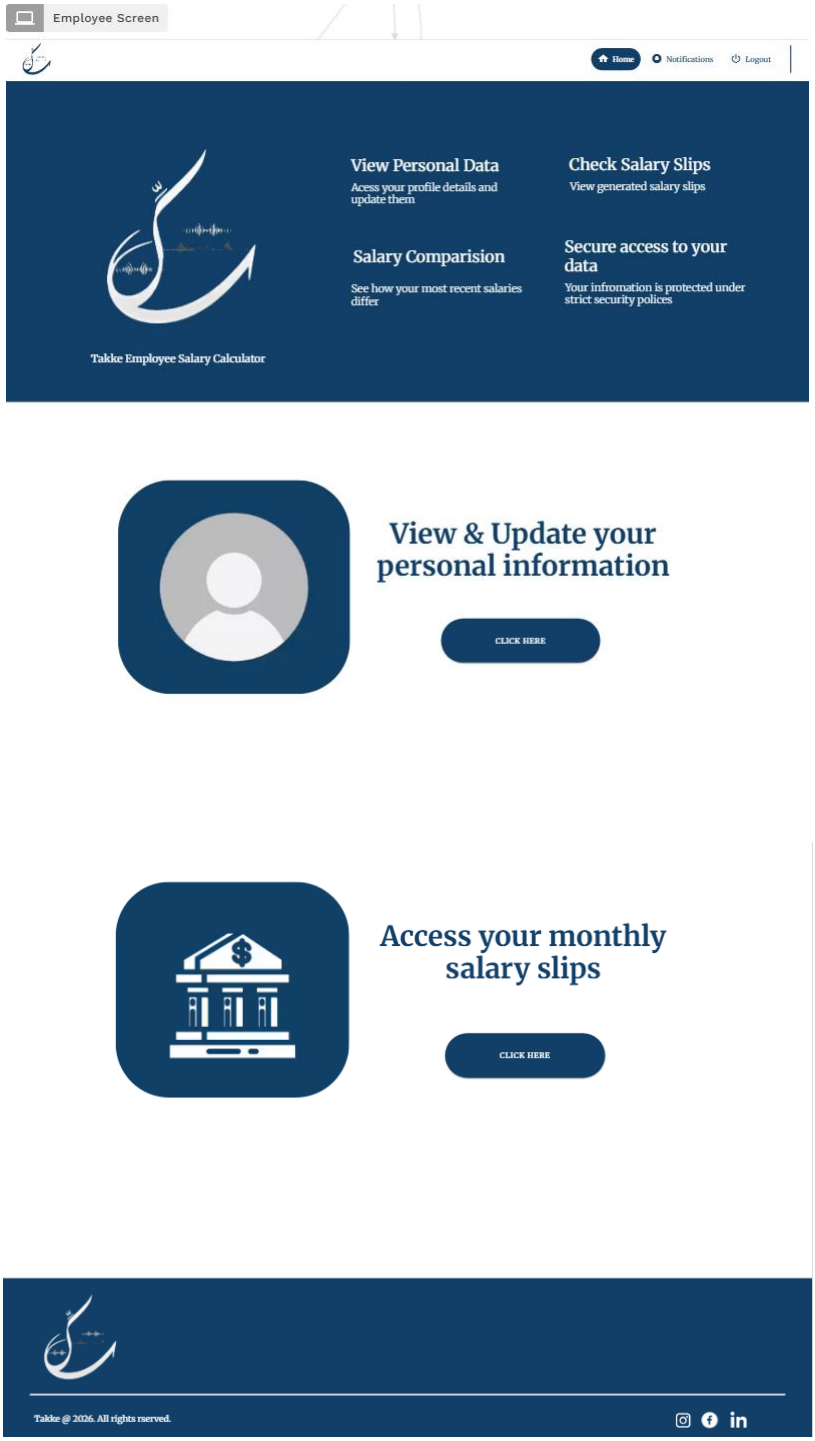

HrEmpSalariesReport
Home
Logout
BACK
First Name Last Name's Salary Slip
Sum
Total Net Salary
Average
Average Net Salary
Sum
Total Deductions
Average
Average Deductions
All Months...
All Years...
Date
Base Salary
Allowances
Deductions
Bonuses
Net Salary
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CalcEmpSalary
BACK
Calculate Salary for EmpId
Month
Select month...
Year
Select year...
Basic Pay
Enter basic pay...
Allowances
Enter allowances...
Bonuses
Enter bonuses...
Deductions
Enter deductions...
CREATE SALARYRECORD

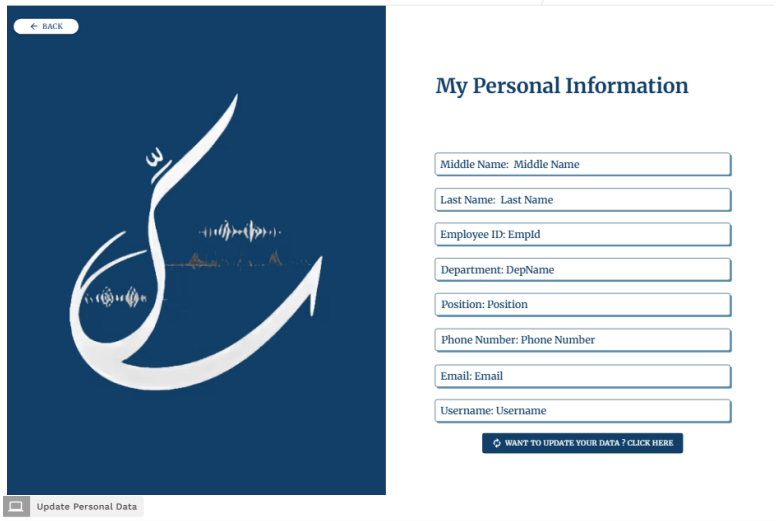

CreateSalaryFeedback
BACK
Salary record for First Name Last Name has Been Updated Successfully
GO HOME
EMPLOYEE LIST


Financial Report
Home
Logout
BACK
Summary Report
Sum
Total Cost
Average
Average Cost
Sum
Total Deductions
Average
Average Deductions
All Months...
All Years...
All Departments ...
Date
Employee ID
Base Salary
Allowances
Deductions
Bonuses
Net Salary
Month / Year
EmpId
Basic Pay
Allowances
Deductions
Bonuses
Net Salary
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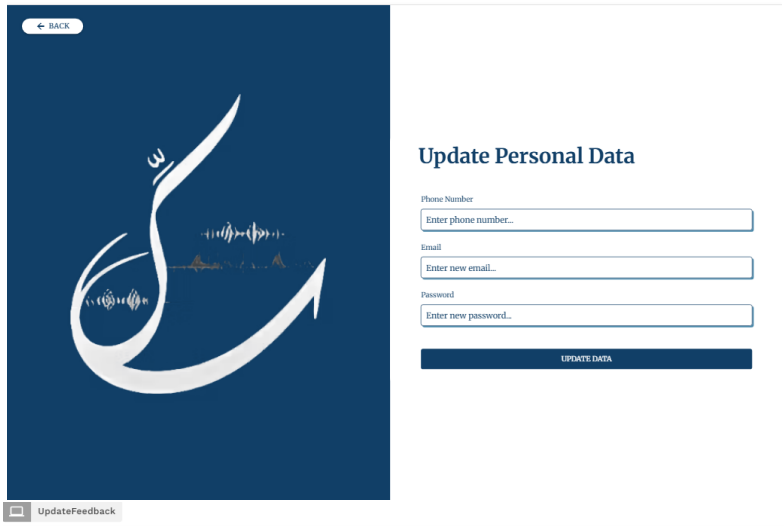
Regular Employee:



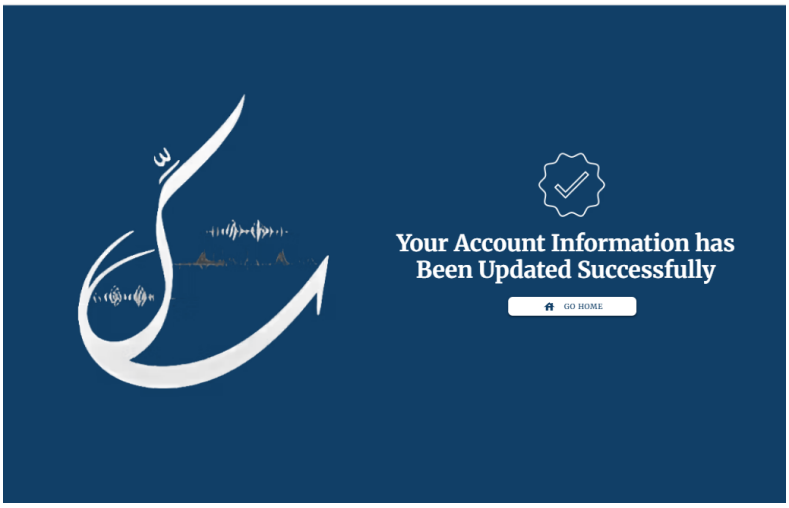
View Personal Data

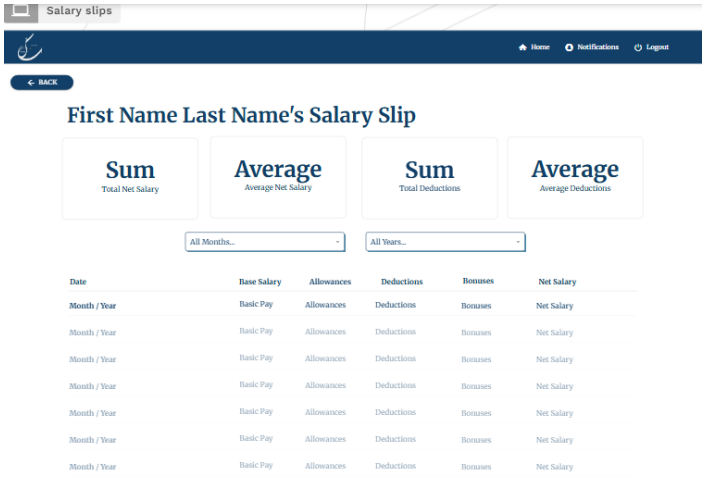


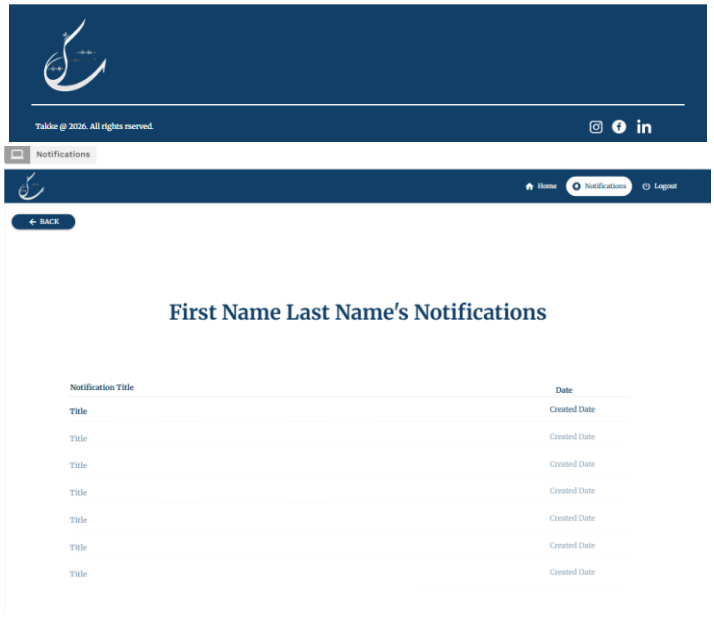
Update Personal Data

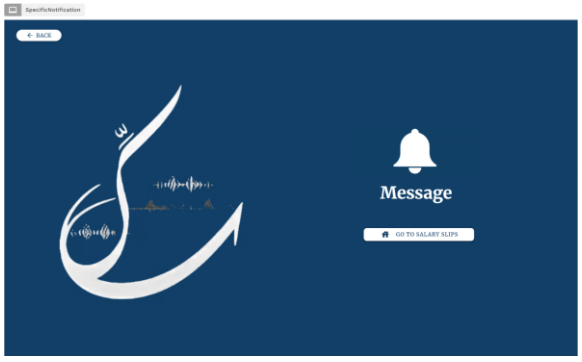


UpdateFeedback




Salary slips
Home
Notifications
Logout
BACK
First Name Last Name's Salary Slip
Sum
Total Net Salary
Average
Average Net Salary
Sum
Total Deductions
Average
Average Deductions
All Months...
All Years...
Date
Base Salary
Allowances
Deductions
Bonuses
Net Salary
Month / Year
Basic Pay
Allowances
Deductions
Bonuses
Net Salary


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Notifications
Home
Notifications
Logout
BACK
First Name Last Name's Notifications
Notification Title
Date
Title
Created Date


SpecificNotification
BACK
Message
GO TO SALARY SLIPS

• Testing

c. Testing plan

In this part, we will create a test plan and implement it in the presented solution for Takke Employee Salary Calculator, as we well test all functional requirements, and 5 non-functional requirements in order to check if the system works well.

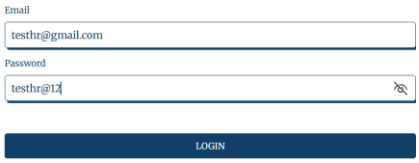
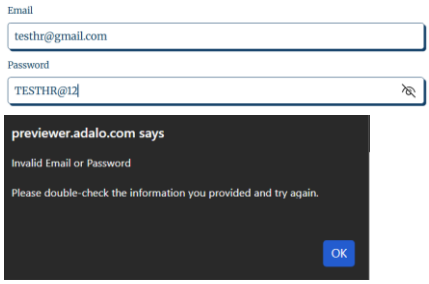
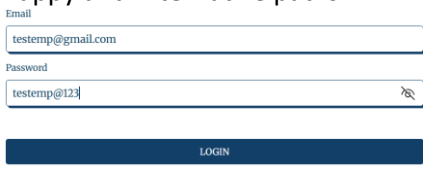
Table 1: Test Cases Plan Table

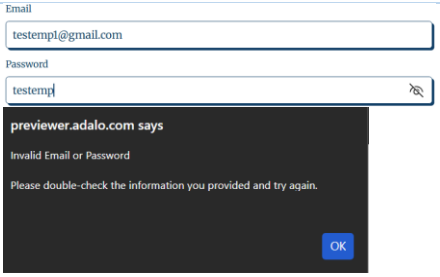
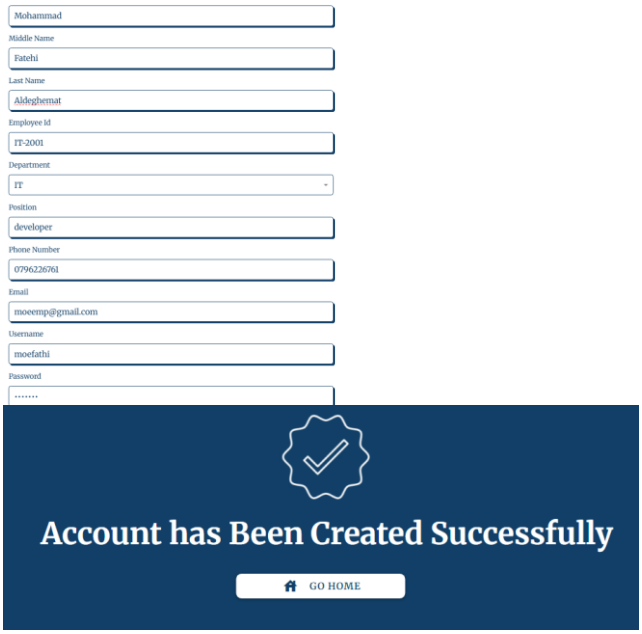
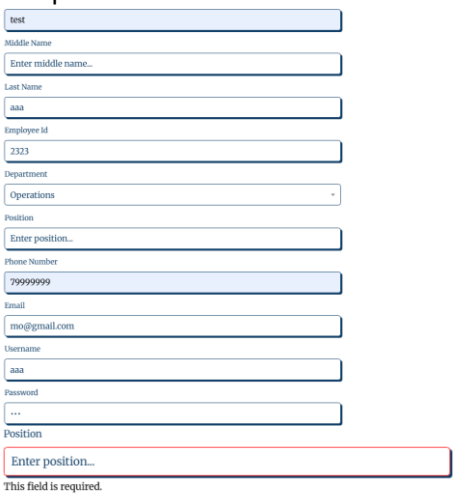
Test ID	Module Name	Description	Test Data	Preconditions	Priority	Scenario
TC-F-A1	Authentication	Login to system based on role	Email: testhr@gmail.com Pass: testhr	Hr account exists in the database	High	HR login -> HR dashboard.
TC-F-A2	Authentication	Login to system based on role	Email: testemp@gmail.com Pass: testemp@123	Employee account exists.	High	HR login -> create employee acc. Employee login -> employee dashboard.
TC-F-H1	HR/Admin Management	HR create employee account	Email: moeemp@gmail.com Pass: moe@123 Username: moefathi empID: IT-2001 First Name: Mohammad Middle Name: Fathi Last Name: Aldeghemat Position: developer Department: IT Phone number: 0796226761	HR user is logged in	High	HR login -> create employee account -> fill values and submit.
TC-F-H2	HR/Admin Management	HR update employee account	For user: moefathi new department: Marketing new position: manager New phone number: 0792032808	HR user is logged in, and employee account exists.	High	HR login -> employees list -> choose employee -> update data -> fill values and submit.
TC-F-H3	HR/Admin Management	HR view a customized employee details report	Filter department: IT	HR user is logged in, and employee exists.	High	HR login -> employees list -> dropdown department.
TC-F-H4	HR/Admin Management	HR view customized employee salary slip report	Filter month: 1 Filter year: 2026	HR user is logged in, and employee and salary records exists.	High	HR login -> employees list -> choose employee -> report -> dropdown month and year.

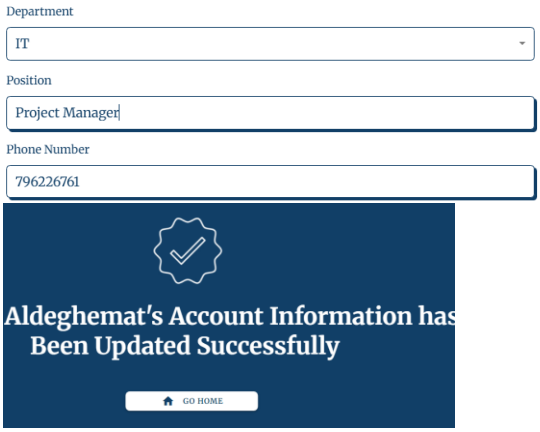
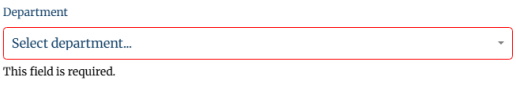
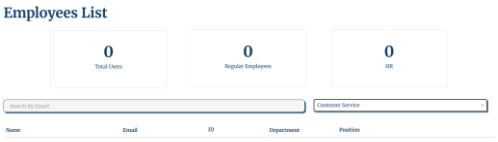
TC-F-H5	HR/Admin Management	HR calculate employee salary	Month: 1 Year: 2026 Basic Pay: 500 Allowances: 50 Bonuses: 100 Deductions: 50	HR user is logged in, and employee exists	High	HR login -> employees list -> calculate salary -> fill values and submit.
TC-F-H6	HR/Admin Management	HR view customized salary slip report	Filter department: IT Filter month: 1 Filter year: 2026	HR user is logged in, and employee and salary records exists.	High	HR login -> salary summary -> dropdown department -> dropdown month -> dropdown year.
TC-F-E1	Employees Management	Employee view personal data	Employee ID: IT-2001	Employee logged in	Medium	Employee login -> view personal data.
TC-F-E2	Employees Management	Employee update personal data	For user: moefathi.d new phone number: 079704080 new email: moefathi@gmail.com new pass: moefathi@123	Employee logged in	Medium	Employee login -> view personal data -> update my data -> fill values and press the submit button
TC-F-E3	Employees Management	Employee receive notifications when salary is calculated	Notification title: your salary has been calculated successfully	HR calculate salary, and employee is logged in	Medium	HR login -> employee list -> calculate salary -> fill values and calculate salary. Emp login -> notifications -> view notification.
TC-F-E4	Employees Management	Employee view customized salary slip report	Month: 12 Year: 2025	Salary record exists, and employee is logged in.	High	Employee login -> salary slip -> dropdown month -> dropdown year
TC-NF-R1	Reliability	Check if report is generated within 5 seconds	Salary records	User is logged in, and salary records exists.	High	Employee login -> start counting time and open salary slip -> compare time.
TC-NF-R2	Reliability	Check if login process is done within 3 seconds	User records	User account exists.	High	Employee fill login values -> start counting time and click on the button -> compare time.

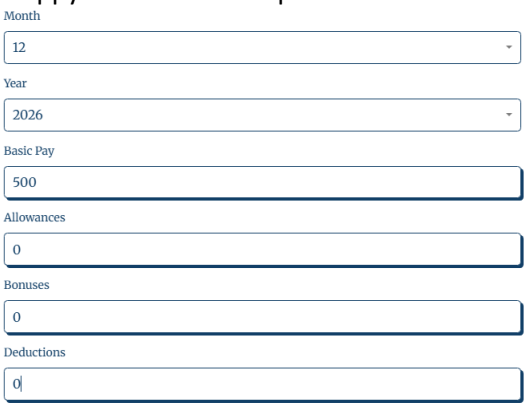
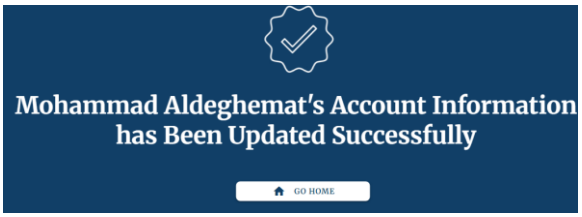
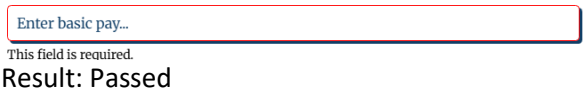
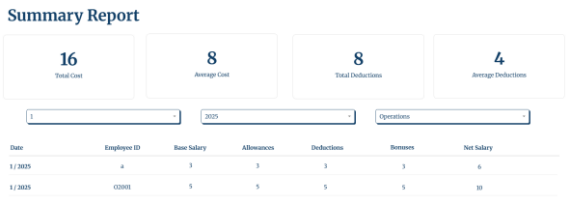
TC-NF-S1	Security	Check if unauthorized access is restricted	User records	Credentials must not exist.	High	user fill login with wrong values -> click button.
TC-NF-P1	Portability	Check if app is accessible on different browsers	Edge, chrome, firefox.	User account exists to access the dashboards.	Medium	Open the application on different browsers.
TC-NF-P2	Portability	Check if app is usable on different screen sizes	Gaming monitor Asus 24', and laptop monitor 15'	User account exists to access the dashboards.	Medium	Open the application on two monitors.

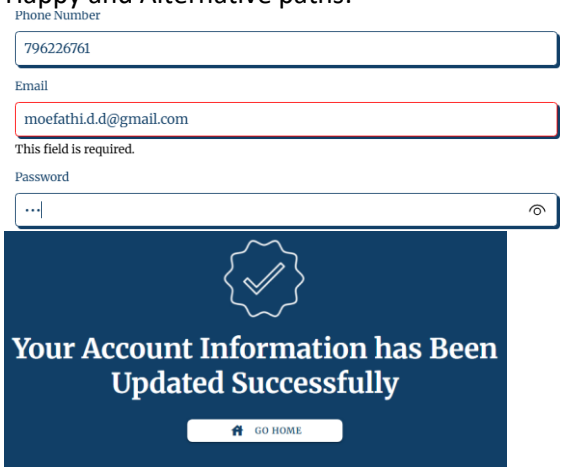
Table 2 Test Cases Execution Table

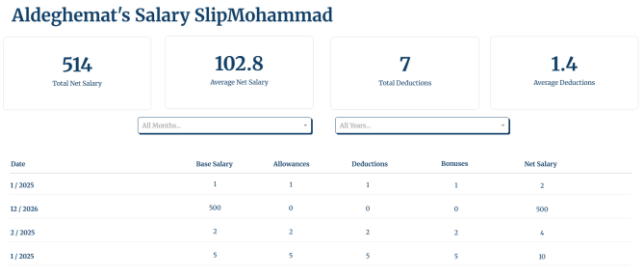
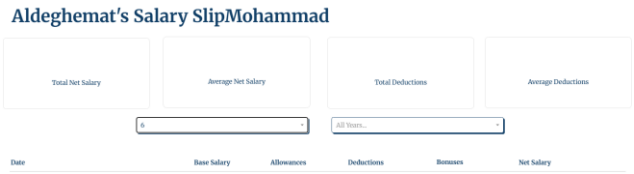
Test id	Steps of Happy path	Steps of Alternative path	Steps of Exceptional path	Expected result	Actual result
TC-F-A1	Enter valid HR email and password.	Enter values with browsers saved credentials (autofill).	Enter wrong email or password.	Happy and Alternative paths: login successfully. Exceptional Path: error.	Happy and Alternative paths:  successful login Exceptional Path:  Result: Passed
TC-F-A2	Enter valid employee email and password.	Refresh page then enters valid employee email and password.	Enter wrong email and password.	Happy and Alternative paths: login successfully. Exceptional Path: error.	Happy and Alternative paths:  successful login Exceptional Path:

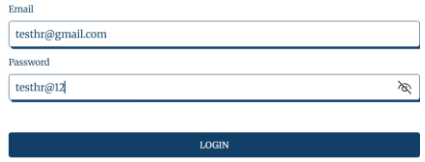
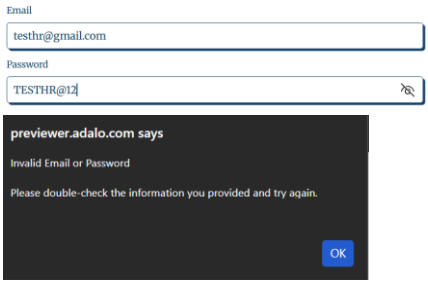
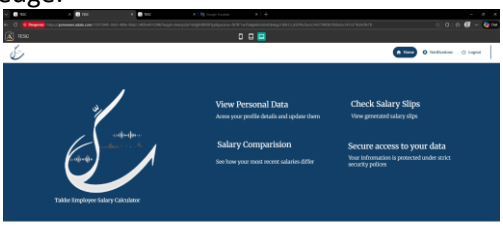
					 Result: Passed
TC-F-H1	HR fills all the fields for one employee.	HR fills all the fields for another employee.	HR leaves fields empty.	Happy and Alternative paths: employee account created successfully Exceptional Path: error.	Happy and Alternative paths:  Exceptional Path:  Result: Passed

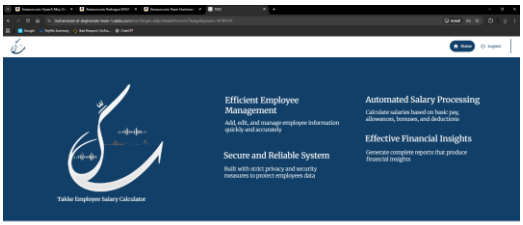
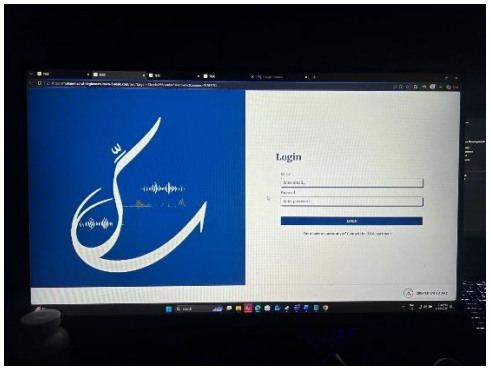
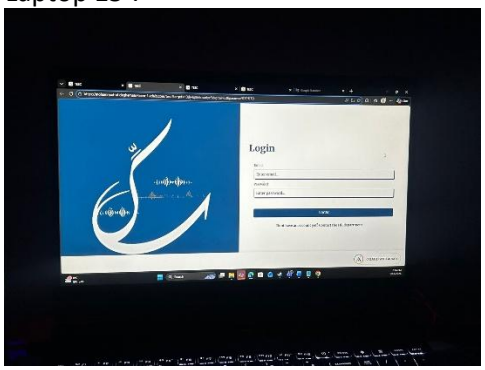
TC-F-H2	HR updates all employee data.	HR update all employee fields with the same old values.	HR leaves some fields empty.	Happy and Alternative paths: employee data updated successfully Exceptional Path: error.	Happy and Alternative paths:  Exceptional Path:  Result: Passed
TC-F-H3	HR filters the employee report by department.	HR view report without filters.	HR select department that has no employees.	Happy and Alternative paths: show IT department employees Exceptional Path: show nothing.	Happy and Alternative paths:  Exceptional Path:  Result: Passed
TC-F-H4	HR filter specific employee salary slip by month and year.	HR filter only month.	HR filters choose future years filters.	Happy and Alternative paths: show salary slip only for a specific month and year Exceptional Path: failed UI	Happy and Alternative paths:  Exceptional Path:

					 Result: Passed
TC-F-H5	HR enter salary values when calculating salaries and press calculate.	HR enters 0 allowances and 0 bonuses.	HR leaves some fields empty.	Happy and Alternative paths: Salary is calculated successfully Exceptional Path: error, the field is required	Happy and Alternative paths:   Exceptional Path:  Result: Passed
TC-F-H6	HR filters summary report by department, month, and year.	HR filter only department and month.	HR filter chooses future years filters	Happy and Alternative paths: Show the report based on department, month, and year. Exceptional Path: failed UI	Happy and Alternative paths:  Exceptional Path:  Result: Passed

TC-F-E1	Employee open personal data page.	Employee open personal data page then update data page then go back to personal data page.	Updating the data with empty fields then open personal data page.	Happy and Alternative paths: Show full data successfully Exceptional Path: The field is required	Happy and Alternative paths:  Exceptional Path:  Result: Passed
TC-F-E2	Employee update all fields and valid data.	Employee update all the fields twice.	Employee leave one field empty.	Happy and Alternative paths: Data updated successfully Exceptional Path: the field is required	Happy and Alternative paths:  Exceptional Path:  Result: Passed
TC-F-E3	Employee open notifications page as soon as a salary calculation then click on the notification	Employee open notifications page after a while from calculating the salary.	Salary is not calculated, so there is no notification for that salary.	Happy and Alternative paths: View all notifications Exceptional Path: no new notification	Happy and Alternative paths: 

	and see the notification content.				Then on click ->  Exceptional Path: No notification Result: Passed
TC-F-E4	Employee select month and year filters for personal salary slips.	Employee do not choose any filter.	Employee choose filter for future month where salary isn't calculated yet.	Happy and Alternative paths: View report based on month and year Exceptional Path: failed UI	Happy and Alternative paths:  Exceptional Path:  Result: Passed
TC-NF-R1	Open any report and it will load within 5 seconds.	Open any report and it will load with exactly 5 seconds.	Open any report and it will load with more than 5 seconds.	Happy and Alternative paths: Report is generated within 5 seconds Exceptional Path: report is generated with more then 5 seconds	Result: Passed The proof will be shown in a separate video.

TC-NF-R2	Login -> dashboard within 3 seconds.	Login -> dashboard with exactly 3 seconds.	Login -> dashboard with more than 3 seconds.	Happy and Alternative paths: Login successfully within 3 seconds Exceptional Path: Login successfully with more than 3 seconds	Result: Passed The proof will be shown in a separate video.
TC-NF-S1	HR authorized user access attempt.	employee authorized user access attempt.	Unauthorized access attempt.	Happy and Alternative paths: Successful login Exceptional Path: error	Happy and Alternative paths:  Then -> successful login Exceptional Path: 
TC-NF-P1	Open and login to the application on chrome then on edge.	Open and login to the application on chrome and edge at the same time.	Use a legacy browser that does not support no-code applications.	Happy and Alternative paths: working well on chrome and edge Exceptional Path: does not work well on legacy browsers	Happy and Alternative paths: edge:  Chrome:

					 Exceptional Path: The exceptional path couldn't be tested as it needed to download a very old version of windows to support very old browser. Result: Passed
TC-NF-P2	Open the application on the asus screen then the laptop screen.	Open the application on the asus screen and the laptop screen at the same time.	Access the application from phone.	Happy and Alternative paths: Work well on supported screen sizes Exceptional Path: failed UI	Happy and Alternative paths: Asus 24':  Laptop 15':  Exceptional Path:

					 Result: Passed